

Constructing asymptotic center–manifold for extended systems

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Abstract

Center-manifold approximation of generic large non-linear system at a multiple co-dimension bifurcation point is considered. The system is perturbed with small parameter perturbations and harmonic forcings so that the system is displaced slightly from the original bifurcation point. We take the “extended systems” approach to analyze such systems which elevates parameter perturbation as well as harmonic forcings as state variables and extends the dimensionality of the center-subspace. The new subspace structure is explicitly evaluated, which is non-trivial when there is resonance between the angular frequencies of the extended modes and the eigenvalues of the unperturbed system. Despite the extension of the operators, It is shown that the construction of an asymptotic center-manifold for such an extended system in fact only requires operations on the original linearized system at bifurcation. A key consequence of taking the “extended systems” perspective is that algorithms for parameter perturbation, harmonic forcing and unperturbed system center-manifolds can be unified into a single algorithm. A Ginzburg-Landau model near bifurcation is used to demonstrate the validity of the approach based on the extended system.

1 Introduction

Center-manifold theory provides an elegant and powerful tool for the analysis of dynamics of systems in the vicinity of the bifurcation point [1, 2, 3, 4, 5, 6, 7]. In such systems the essential dynamics are primarily driven by the dynamics in the center-manifold, which is a continuation of the center space of the linearized system into the non-linear regime. The center manifold has the same dimension as the center space leading to a low-dimensional representation of what might originally be a very high dimensional system. The ideas underlying the theory are powerful and extend beyond center manifolds, having found applicability to general invariant manifolds [8, 9]. In the hydrodynamic literature a similar approach has proliferated under the general framework of

“weakly non-linear analysis” [10, 11, 12, 13], although analytical work utilizing ideas of scale separation have been undertaken much earlier as well [14, 15, 16]. Fujimura [17] has claimed that the two methods (center-manifold and weakly non-linear analysis) are effectively equivalent although, the underlying ideas of the two methods are somewhat distinct.

While the importance of invariant manifold in general, and center-manifolds in particular, have long been recognized, its application was initially largely confined to small ordinary differential equation type systems [1, 6, 5]. There has however been a steady progress towards application of invariant manifold theory to larger and more complex systems. The work of Coullet and Spiegel [20] is seminal in this regard which presented a method

to directly evaluate center-manifold approximations (in their normal form) of partial differential equation type problems. The authors then applied the method to the hydrodynamic problem of double diffusion. The approach has been strongly advocated by Roberts [9] and largely underpins the algorithmic procedure presented in [21].

Subsequently, the work of Cabré et al. [22, 23, 24], Haro et al. [25] introduced the parametrization method which has been successfully utilized for the calculation of spectral sub-manifolds (SSM) [26, 27, 28]. The framework is quite general and allows for any particular style of representation of the reduced dynamics. Indeed the normal-form evaluation procedure of [20] can be seen as a particular choice of parameterization. As a side note, the essential idea of imposing an arbitrary choice of parameterization to the invariant manifold and evaluating the corresponding asymptotic vectors was also presented by [19], albeit via more specific examples.

Often one is interested in cases where the system is perturbed slightly away from the bifurcation point or through the application of small amplitude time dependent forcing. This leads us to the territory of non-autonomous dynamical systems. Theoretical work on invariant manifolds of general non-autonomous systems exists [29, 30, 31, 32], however, practical applications to large systems remains few and far in between. The limited applications on small systems may often rely on integral transforms requiring convolutions in time [18, 30, 32]. Such operations tend to impose prohibitive computational costs for large systems.

A simpler scenario exists when the time-dependence of the system can be specified with a small number of harmonics. The work presented in [26, 27, 28] leverages the parameterization method to develop a methodology for the computation of invariant manifolds of autonomous and non-autonomous dynamical systems. An alternate approach is to elevate parameter perturbations and harmonic forcings to state variables. This of course is not a novel idea and is routinely performed “as a trick” for small dynamical systems, with some examples of application to larger systems [33, 18]. However, a systematic examination of such an operation for general dynamical systems and its consequences for asymptotic evaluations of invariant manifolds is lacking. The current

work investigates this in a general setting, illuminating the structure of the new tangent spaces and subsequent local extended center-manifold.

As it turns out this detailed examination reveals a few pleasant surprises. Despite a non-trivial expansion of the center subspace, high order asymptotic terms of the local manifold can be evaluated using the linear operators of the original (unextended) system. This is an important aspect for large scale systems since solution strategies for such systems often rely on exploiting the inherent structure of the system. Changing the system matrices due to the expansion of the state vector could potentially destroy the structure of the operators causing existing algorithms to fail. Fortunately, the system matrices require no modification. Secondly, since the external forcing and parameter perturbations are treated at par with internal modes, the algorithmic treatment is unified. A single algorithm therefore incorporates all the three cases of center-manifold evaluations - unperturbed system, parameter perturbations and harmonically forced systems.

A more subtle difference arises with the interpretation of the invariant manifold in the non-autonomous setting. The approach in [26] is to consider the tori resulting from the deformation of the hyperbolic fixed point of the original system under small amplitude harmonic forcing, and evaluate the phase dependent invariant manifold attached to this tori. However, as pointed out in [26], the tori is not guaranteed to exist for a non-hyperbolic fixed point. A center-manifold approximation is the appropriate setting in such a case. The tori (should they exist) are simply part of the larger center-manifold emanating from the fixed point of the extended system, which is guaranteed to exist by the center-manifold theorem.

The current work therefore can be considered complementary to that of [26]. It presents in a general setting the treatment of parameter perturbation, something that has been treated by way of example of a perturbed Lorenz system in [26]. In addition, it allows for the construction of center-manifolds of non-autonomous systems even in the case of non-hyperbolic fixed points and unifies the algorithmic treatment for all cases.

The remainder of the manuscript is laid out as follows. Section 2 presents the problem in a general setting and subsequently evaluates the structure

of the extended tangent spaces under parameter perturbation and harmonic forcings. Section 4 then builds the algorithms for the construction of asymptotic center-manifolds of the extended systems while exploiting the tangent space structure derived earlier. The presented algorithm is then applied to a Ginzburg-Landau model under various conditions of system extension in section 5. Concluding remarks are made in section 6.

2 Problem Definition

2.1 The Standard System

Consider a system with its state variable denoted by $\mathbf{u}$, consisting of p parameters given by the vector $\boldsymbol{\mu}$, and an evolution law given by

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{f}(\mathbf{u}, \boldsymbol{\mu}). \quad (1)$$

The system state $\mathbf{u}$ is N -dimensional where, N is assumed to be large. The evolution law may arise from a finite set of ordinary differential equations or, may be a finite representation of an infinite dimensional partial differential equation as often arises from the numerical discretization of the space-time evolution equations of physical systems. Typically, physical systems will have real parameters $\boldsymbol{\mu}$ however, allowing for complex parameters presents no additional difficulty and we assume that $\boldsymbol{\mu} \in \mathbb{C}^p$. Without loss of generality one may assume $\mathbf{u} = \mathbf{u}^b$ to be a fixed point of the system for a certain set of parameters $\boldsymbol{\mu} = \boldsymbol{\mu}^b$, *i.e.* ,

$$\mathbf{f}(\mathbf{u}^b, \boldsymbol{\mu}^b) = \mathbf{0}. \quad (2)$$

We may consider the evolution of the deviation from the fixed point solution $\mathbf{u} = \mathbf{u} - \mathbf{u}^b$, and divide the evolution operator into its linear and non-linear parts as,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b) \quad (3)$$

where,

$$\mathbf{L} = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right]_{(\mathbf{u}=\mathbf{u}^b, \boldsymbol{\mu}=\boldsymbol{\mu}^b)} \\ \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b) = \mathbf{f}(\mathbf{u}^b + \mathbf{u}, \boldsymbol{\mu}^b) - \mathbf{L}\mathbf{u}.$$

Furthermore, it is assumed that at $\boldsymbol{\mu} = \boldsymbol{\mu}^b$ the system undergoes a multiple co-dimension bifurcation and that the linearized operator $\mathbf{L} \in \mathbb{C}^{N \times N}$ has a discrete set of n critical eigenvalues λ_i^C , $i \in 1, \dots, n$, with zero growth rate ($\Re(\lambda_i^C) = 0$). This includes both purely real critical eigenvalues with $\Re(\lambda_i^C) = 0$, $\Im(\lambda_i^C) = 0$, and purely imaginary critical eigenvalues with $\Re(\lambda_i^C) = 0$, $\Im(\lambda_i^C) \neq 0$. Accordingly, the system has n critical eigenpairs $\{\lambda_i^C, \mathbf{v}_i\}$, $i \in 1 \dots n$, where $\mathbf{v}_i$ represents the eigenvector corresponding to λ_i^C . For real systems the purely imaginary critical eigenvalues will occur in conjugate pairs.

The adjoint of the direct linearized operator $\mathbf{L}$ is denoted as $\mathbf{L}^\dagger$ which also has n critical eigenpairs $\{\lambda_i^{C*}, \mathbf{w}_i\}$. The scaling of the eigenvectors of the direct and adjoint operators may be defined such that they satisfy the biorthogonality condition

$$\langle \mathbf{w}_i, \mathbf{v}_j \rangle = \mathbf{w}_i^\dagger \mathbf{v}_j = \delta_{i,j}, \quad (4)$$

where $\langle \cdot, \cdot \rangle$ is the standard inner-product and $\delta_{i,j}$ is the Kronecker-delta function. The superscript † is used here to indicate a complex-conjugated transpose of vectors and matrices. For scalars complex-conjugation is indicated by a $*$ superscript. The superscript H which is often used for the same purpose is reserved for objects referring to harmonic forcings. One may define the matrices $\mathbf{V}$ and $\mathbf{W}$, whose columns comprise of the n critical eigenvectors of $\mathbf{L}$ and $\mathbf{L}^\dagger$ as,

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n], \quad (5a)$$

$$\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n]. \quad (5b)$$

Due to the chosen scaling denoted by equation (4) the matrices satisfy,

$$\mathbf{W}^\dagger \mathbf{V} = \mathbf{I}_n, \quad (6)$$

where, $\mathbf{I}_n$ represents an identity matrix of order $n \times n$. The columns of the matrices $\mathbf{V}$ and $\mathbf{W}$ span the n -dimensional center-subspace of the linear operators $\mathbf{L}$ and $\mathbf{L}^\dagger$ respectively.

Additionally, $\boldsymbol{\Lambda}_C$ is defined as an $n \times n$ diagonal matrix consisting of the critical eigenvalues of $\mathbf{L}$, with the implicit assumption of diagonalizability of $\mathbf{L}$ within the center subspace. The ordering of the eigenvalues in $\boldsymbol{\Lambda}_C$ and eigenvectors in $\mathbf{V}$ and $\mathbf{W}$ is consistent with each other so that we have

the eigenvalue relations,

$$\mathbf{L}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}_C, \quad \mathbf{L}^\dagger\mathbf{W} = \mathbf{W}\mathbf{\Lambda}_C^\dagger. \quad (7)$$

2.2 From the Standard System to the Extended System

The same system is now considered for the case of parameter perturbation so that the evolution dynamics are now evaluated at parameter values of $\boldsymbol{\mu} = \boldsymbol{\mu}^b + \boldsymbol{\nu}$, with $\boldsymbol{\nu}$ being the vector of parameter perturbations, $\boldsymbol{\nu} = [\nu_1; \nu_2; \dots; \nu_p]$. In addition, the system is forced with harmonic forcing of the type,

$$\mathbf{f}^H(\boldsymbol{\theta}) = \sum_{i=1}^h \mathbf{f}_i^H \theta_i(t); \quad d\theta_i/dt = \lambda_i^H \theta_i, \quad (8)$$

where, $\boldsymbol{\theta} = [\theta_1; \theta_2; \dots; \theta_h]$ and the various θ_i are treated as dynamic variables. λ_i^H is purely imaginary, with the imaginary part representing the angular frequency of the forcing. The vector of angular frequencies is denoted as $\boldsymbol{\lambda}^H = [\lambda_1^H; \lambda_2^H; \dots; \lambda_h^H]$. Constant forcing terms are readily incorporated when a particular (or multiple) $\lambda_i^H = 0$. The (time-independent) shape of the i^{th} harmonic forcing is denoted as $\mathbf{f}_i^H$. Denoting the new harmonically forced evolution operator as,

$$\mathbf{f}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) = \mathbf{f}(\mathbf{u}^b + \mathbf{u}, \boldsymbol{\mu}^b + \boldsymbol{\nu}) + \mathbf{f}^H(\boldsymbol{\theta}), \quad (9)$$

one may again separate out the linear and non-linear parts as,

$$\begin{aligned} \mathbf{L} &= \left[\frac{\partial \mathbf{f}'}{\partial \mathbf{u}} \right]_{(\mathbf{u}=\mathbf{0}, \boldsymbol{\nu}=\mathbf{0}, \boldsymbol{\theta}=\mathbf{0})} = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right]_{(\mathbf{u}=\mathbf{u}^b, \boldsymbol{\mu}=\boldsymbol{\mu}^b)}, \\ \mathbf{L}_\nu &= \left[\frac{\partial \mathbf{f}'}{\partial \boldsymbol{\nu}} \right]_{(\mathbf{u}=\mathbf{0}, \boldsymbol{\nu}=\mathbf{0}, \boldsymbol{\theta}=\mathbf{0})} = \left[\frac{\partial \mathbf{f}}{\partial \boldsymbol{\mu}} \right]_{(\mathbf{u}=\mathbf{u}^b, \boldsymbol{\mu}=\boldsymbol{\mu}^b)}, \\ \mathbf{L}_\theta &= \left[\frac{\partial \mathbf{f}'}{\partial \boldsymbol{\theta}} \right]_{(\mathbf{u}=\mathbf{0}, \boldsymbol{\nu}=\mathbf{0}, \boldsymbol{\theta}=\mathbf{0})} = \left[\frac{\partial \mathbf{f}^H}{\partial \boldsymbol{\theta}} \right]_{(\boldsymbol{\theta}=\mathbf{0})}, \end{aligned}$$

$$\mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) = \mathbf{f}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) - \mathbf{L}\mathbf{u} - \mathbf{L}_\nu \boldsymbol{\nu} - \mathbf{L}_\theta \boldsymbol{\theta}.$$

where, $\mathbf{L} \in \mathbb{C}^{N \times N}$, $\mathbf{L}_\nu \in \mathbb{C}^{N \times p}$, $\mathbf{L}_\theta \in \mathbb{C}^{N \times h}$ which results in the evolution equation given by,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{L}\mathbf{u} + \mathbf{L}_\nu \boldsymbol{\nu} + \mathbf{L}_\theta \boldsymbol{\theta} + \mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}).$$

The diagonal matrix for the harmonic angular frequencies is represented by $\mathbf{\Lambda}_H = \text{diagm}(\boldsymbol{\lambda}^H)$. Similarly, $\boldsymbol{\lambda}^P = [\lambda_1^P; \lambda_2^P; \dots; \lambda_p^P]$ is defined as a vector of length p , with all $\lambda_i^P = 0$ identically, for $i \in 1, \dots, p$. These may be thought of parameter eigenvalues or parameter angular frequencies depending on the point of view. $\mathbf{\Lambda}_P = \text{diagm}(\boldsymbol{\lambda}^P)$ is defined as an $p \times p$ null matrix, to represent the trivial linear evolution matrix for the parameter variables $\boldsymbol{\nu}$. The extended system including the evolution equations for the newly introduced variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$, henceforth referred to as the *extended variables*, is given by,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{L}\mathbf{u} + \mathbf{L}_\nu \boldsymbol{\nu} + \mathbf{L}_\theta \boldsymbol{\theta} + \mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) \quad (10a)$$

$$\frac{d\boldsymbol{\nu}}{dt} = \mathbf{\Lambda}_P \boldsymbol{\nu} \quad (10b)$$

$$\frac{d\boldsymbol{\theta}}{dt} = \mathbf{\Lambda}_H \boldsymbol{\theta}, \quad (10c)$$

An overhead hat notation, $\widehat{\cdot}$, is used to refer to variables and operators associated with the extended space, so that the new state vector, non-linear term and the linearized direct and adjoint operators are denoted by $\widehat{\mathbf{u}}$, $\widehat{\mathcal{N}'}$, $\widehat{\mathbf{L}}$ and $\widehat{\mathbf{L}}^\dagger$, that are defined as,

$$\widehat{\mathbf{u}} = [\mathbf{u}; \boldsymbol{\nu}; \boldsymbol{\theta}], \quad \widehat{\mathcal{N}'} = [\mathcal{N}'; \mathbf{0}; \mathbf{0}],$$

$$\widehat{\mathbf{L}} = \begin{bmatrix} \mathbf{L} & \mathbf{L}_\nu & \mathbf{L}_\theta \\ \mathbf{0} & \mathbf{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Lambda}_H \end{bmatrix}, \quad \widehat{\mathbf{L}}^\dagger = \begin{bmatrix} \mathbf{L}^\dagger & \mathbf{0} & \mathbf{0} \\ \mathbf{L}_\nu^\dagger & \mathbf{\Lambda}_P^\dagger & \mathbf{0} \\ \mathbf{L}_\theta^\dagger & \mathbf{0} & \mathbf{\Lambda}_H^\dagger \end{bmatrix}. \quad (11)$$

The full system is now M -dimensional, with $M = N + p + h$, and may be represented compactly as,

$$\frac{\partial \widehat{\mathbf{u}}}{\partial t} = \widehat{\mathbf{L}}\widehat{\mathbf{u}} + \widehat{\mathcal{N}'}(\widehat{\mathbf{u}}). \quad (12)$$

The system defined by equation (10), or its compact form in equation (12), with additional dynamical equations for the extended variables will be referred to as the *extended system*.

3 Extended Subspace Structure

The extended system has a larger center subspace of dimension $m = n + p + h$, as compared to the original system due to the $p + h$ additional variables introduced. While the extension of the system seems almost trivial, the eigenvector structure in the new dimensions however is far from trivial. Nevertheless, the extended eigenspace structure can be deduced directly from the properties of the original system. It is useful to examine the structure of the extended subspace since the adjoint extended critical subspace does have a special structure which has important practical implications for the asymptotic evaluation of the center-manifold of the extended system.

There are however some subtleties involved when the system extension introduces new eigenvalues that are resonant with the original linear system $\mathbf{L}$. In such a scenario the system extension leads to a generalized eigenspace for the resonant eigenvalues. This is demonstrated in the next subsections where the subspace structure is evaluated in detail for both the resonant and non-resonant cases. This is first done for the extended operator $\widehat{\mathbf{L}}$ and then for the adjoint operator $\widehat{\mathbf{L}}^\dagger$.

3.1 Subspace structure for $\widehat{\mathbf{L}}$

Observe that if $\{\lambda_i^C, \mathbf{v}_i\}$ is a critical eigenpair of the original linearized operator $\mathbf{L}$, then its extension to $\widehat{\mathbf{L}}$ satisfies,

$$\widehat{\mathbf{v}}_i^C = [\mathbf{v}_i; \mathbf{0}; \mathbf{0}], \quad (13a)$$

$$(\lambda_i^C \widehat{\mathbf{I}} - \widehat{\mathbf{L}}) \widehat{\mathbf{v}}_i^C = \mathbf{0}, \quad (13b)$$

$\forall i \in 1, \dots, n$, which may be verified via simple substitution. Here $\widehat{\mathbf{I}}$ is the identity matrix for the extended system. Hence all critical eigenmodes of $\mathbf{L}$ can be trivially extended to $\widehat{\mathbf{L}}$. This definition for the extension of the eigenvectors in fact applies to *all* eigenmodes of $\mathbf{L}$, not just the critical ones.

The additional critical eigenpairs introduced due to system extension need to be deduced. The new eigenmodes introduced due to parameter perturbations will be referred to as the *parameter* modes while those generated due to the harmonic forcings will be referred to as *forcing* modes. Accordingly, two sets of canonical unit vectors are

defined such that,

$$\mathbf{e}_i^P = \underbrace{[0; 0; \dots; 0]_{i-1}}_{i-1}; \underbrace{[1; 0; 0; \dots; 0]_{p-i}}_{p-i}, \quad (14a)$$

$$\mathbf{e}_i^H = \underbrace{[0; 0; \dots; 0]_{i-1}}_{i-1}; \underbrace{[1; 0; 0; \dots; 0]_{h-i}}_{h-i}. \quad (14b)$$

If the original linear system $\mathbf{L}$ is non-singular then, the structure of the p parameter eigenpairs is given by $\{\lambda_i^P, \widehat{\mathbf{v}}_i^P\}$, which satisfy

$$\widehat{\mathbf{v}}_i^P = [\mathbf{v}_i^P; \mathbf{e}_i^P; \mathbf{0}], \quad \forall i \in 1, \dots, p, \quad (15a)$$

$$(\lambda_i^P \mathbf{I} - \mathbf{L}) \mathbf{v}_i^P = \mathbf{L}_\nu \mathbf{e}_i^P, \quad (\lambda_i^P = 0). \quad (15b)$$

It can be verified that $(\lambda_i^P \widehat{\mathbf{I}} - \widehat{\mathbf{L}}) \widehat{\mathbf{v}}_i^P = \mathbf{0}$.

However, if $\mathbf{L}$ is singular then equation (15) is indeterminate and the definition needs to be modified. For the i^{th} parameter mode, define a vector $\boldsymbol{\gamma}_i^P \in \mathbb{C}^n$, where the j^{th} element of the vector is given by,

$$(\boldsymbol{\gamma}_i^P)_j = \begin{cases} \mathbf{w}_j^\dagger \mathbf{L}_\nu \mathbf{e}_i^P, & \text{if } \lambda_i^P = \lambda_j^C, \\ 0, & \text{otherwise,} \end{cases} \quad (16)$$

and we have a new definition of $\widehat{\mathbf{v}}_i^P$ satisfying,

$$\widehat{\mathbf{v}}_i^P = [\mathbf{v}_i^P; \mathbf{e}_i^P; \mathbf{0}], \quad \forall i \in 1, \dots, p, \quad (17a)$$

$$(\lambda_i^P \mathbf{I} - \mathbf{L}) \mathbf{v}_i^P = \mathbf{L}_\nu \mathbf{e}_i^P - \mathbf{V} \boldsymbol{\gamma}_i^P, \quad (\lambda_i^P = 0), \quad (17b)$$

$$\mathbf{w}_j^\dagger \mathbf{v}_i^P = 0, \quad \text{if } \lambda_i^P = \lambda_j^C. \quad (17c)$$

In the singular case, $\widehat{\mathbf{v}}_i^P$ is no longer an eigenvector of $\widehat{\mathbf{L}}$, however, it is a generalized eigenvector satisfying $(\lambda_i^P \widehat{\mathbf{I}} - \widehat{\mathbf{L}})^2 \widehat{\mathbf{v}}_i^P = \mathbf{0}$. Note that $\boldsymbol{\gamma}_i^P$ is identically zero for the non-singular case and equation (17) reduces to equation (15). Henceforth the parameter modes will be defined by equation (17), since it is the more general definition.

Similarly, when the harmonic forcing is non-resonant, the h forcing modes, denoted as $\{\lambda_i^H, \widehat{\mathbf{v}}_i^H\}$, generated by system extension due to $\boldsymbol{\theta}$ have the following structure,

$$\widehat{\mathbf{v}}_i^H = [\mathbf{v}_i^H; \mathbf{0}; \mathbf{e}_i^H], \quad \forall i \in 1, \dots, h, \quad (18a)$$

$$(\lambda_i^H \mathbf{I} - \mathbf{L}) \mathbf{v}_i^H = \mathbf{L}_\theta \mathbf{e}_i^H. \quad (18b)$$

One may again verify $(\lambda_i^H \hat{\mathbf{I}} - \hat{\mathbf{L}}) \hat{\mathbf{v}}_i^H = \mathbf{0}$. On the other hand, when the harmonic forcing is resonant with some of the angular frequencies of $\mathbf{L}$ then, analogous to the case of parameter modes, one defines,

$$(\gamma_i^H)_j = \begin{cases} \mathbf{w}_j^\dagger \mathbf{L}_\theta \mathbf{e}_i^H, & \text{if } \lambda_i^H = \lambda_j^C \\ 0, & \text{otherwise,} \end{cases} \quad (19)$$

where, $(\gamma_i^H)_j$ is the j^{th} component of the vector γ_i^H . The extended modes are then given by,

$$\hat{\mathbf{v}}_i^H = [\mathbf{v}_i^H; \mathbf{0}; \mathbf{e}_i^H], \quad \forall i \in 1, \dots, h. \quad (20a)$$

$$(\lambda_i^H \mathbf{I} - \mathbf{L}) \mathbf{v}_i^H = \mathbf{L}_\theta \mathbf{e}_i^H - \mathbf{V} \gamma_i^H, \quad (20b)$$

$$\mathbf{w}_j^\dagger \mathbf{v}_i^H = 0, \quad \text{if } \lambda_i^H = \lambda_j^C, \quad (20c)$$

In the resonant case, this satisfies the generalized eigenvalue equation $(\lambda_i^H \hat{\mathbf{I}} - \hat{\mathbf{L}})^2 \hat{\mathbf{v}}_i^H = \mathbf{0}$. Again, equation (20) reduces to equation (18) when the forcing is non-resonant. Henceforth equation (20) will be used as the definition of the forcing modes. The two sets of vectors γ_i^P and γ_i^H are represented together as matrices,

$$\mathbf{\Gamma}_P = [\gamma_1^P, \gamma_2^P, \dots, \gamma_p^P], \quad (21a)$$

$$\mathbf{\Gamma}_H = [\gamma_1^H, \gamma_2^H, \dots, \gamma_h^H], \quad (21b)$$

The critical eigenspace of $\hat{\mathbf{L}}$ can be represented by three matrices

$$\hat{\mathbf{V}}_C = \begin{bmatrix} \mathbf{V} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \hat{\mathbf{V}}_P = \begin{bmatrix} \mathbf{V}_P \\ \mathbf{I}_p \\ \mathbf{0} \end{bmatrix}, \quad \hat{\mathbf{V}}_H = \begin{bmatrix} \mathbf{V}_H \\ \mathbf{0} \\ \mathbf{I}_h \end{bmatrix}, \quad (22)$$

where, $\mathbf{V}$ is the matrix of the critical eigenvectors of $\mathbf{L}$. The matrix $\mathbf{V}_P$ comprises of the p vectors $\mathbf{v}_i^P$ of the parameter modes, and, $\mathbf{V}_H$ comprises of the h vectors $\mathbf{v}_i^H$ of the forcing modes. The matrix of vectors of the extended eigenspace is denoted by, $\hat{\mathbf{V}} = [\hat{\mathbf{V}}_C, \hat{\mathbf{V}}_P, \hat{\mathbf{V}}_H]$ with the individual vectors denoted as $\hat{\mathbf{v}}_i$, $i \in 1, 2, \dots, m$.

3.2 Subspace structure for $\hat{\mathbf{L}}^\dagger$

The eigenspace structure for the adjoint problem is important to consider since these vectors define

the projection operators into the extended center subspace. The critical eigenmodes of the $\hat{\mathbf{L}}^\dagger$ can be extended via the following definition,

$$\hat{\mathbf{w}}_i^C = [\mathbf{w}_i; \zeta_i^P; \zeta_i^H], \quad (23a)$$

$$(\lambda_i^{C*} \mathbf{I} - \mathbf{L}^\dagger) \mathbf{w}_i = 0, \quad \forall i \in 1, \dots, n, \quad (23b)$$

with vectors ζ_i^P and ζ_i^H defined such that the j^{th} component of ζ_i^P given by,

$$(\zeta_i^P)_j = \begin{cases} (\mathbf{L}_\nu^\dagger \mathbf{w}_i)_j / (\lambda_i^{C*} - \lambda_j^{P*}); & \text{if } \lambda_i^{C*} \neq \lambda_j^{P*}, \\ 0; & \text{if } \lambda_i^{C*} = \lambda_j^{P*}, \end{cases} \quad (24)$$

$\forall j \in 1, \dots, p$, and, the j^{th} component of ζ_i^H is given by,

$$(\zeta_i^H)_j = \begin{cases} (\mathbf{L}_\theta^\dagger \mathbf{w}_i)_j / (\lambda_i^{C*} - \lambda_j^{H*}); & \text{if } \lambda_i^{C*} \neq \lambda_j^{H*}, \\ 0; & \text{if } \lambda_i^{C*} = \lambda_j^{H*}, \end{cases} \quad (25)$$

$\forall j \in 1, \dots, h$. Here, $(\mathbf{L}_\nu^\dagger \mathbf{w}_i)_j$ refers to the j^{th} component of the vector $(\mathbf{L}_\nu^\dagger \mathbf{w}_i)$, and likewise for $(\mathbf{L}_\theta^\dagger \mathbf{w}_i)_j$. It may be verified that the definition is consistent with the eigenvalue equation, $(\lambda_i^{C*} \hat{\mathbf{I}} - \hat{\mathbf{L}}^\dagger) \hat{\mathbf{w}}_i^C = 0$, for the non-resonant cases, and satisfies generalized eigenvalue equation, $(\lambda_i^{C*} \hat{\mathbf{I}} - \hat{\mathbf{L}}^\dagger)^2 \hat{\mathbf{w}}_i^C = 0$, for the resonant cases.

The adjoint parameter modes are given simply by,

$$\hat{\mathbf{w}}_i^P = [\mathbf{0}; \mathbf{e}_i^P; \mathbf{0}], \quad \forall i \in 1, \dots, p, \quad (26)$$

which satisfies $(\lambda_i^{P*} \hat{\mathbf{I}} - \hat{\mathbf{L}}^\dagger) \hat{\mathbf{w}}_i^P = 0$. And the adjoint forcing modes are given by,

$$\hat{\mathbf{w}}_i^H = [\mathbf{0}; \mathbf{0}; \mathbf{e}_i^H], \quad \forall i \in 1, \dots, h, \quad (27)$$

which satisfies $(\lambda_i^{H*} \hat{\mathbf{I}} - \hat{\mathbf{L}}^\dagger) \hat{\mathbf{w}}_i^H = 0$.

The extended eigenspace of $\hat{\mathbf{L}}^\dagger$ can be represented with matrices as

$$\hat{\mathbf{W}}_C = \begin{bmatrix} \mathbf{W} \\ \mathbf{Z}_P \\ \mathbf{Z}_H \end{bmatrix}, \quad \hat{\mathbf{W}}_P = \begin{bmatrix} \mathbf{0} \\ \mathbf{I}_p \\ \mathbf{0} \end{bmatrix}, \quad \hat{\mathbf{W}}_H = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{I}_h \end{bmatrix}, \quad (28)$$

where, $\mathbf{W}$ is the matrix of the adjoint critical eigenvectors of $\mathbf{L}^\dagger$. The matrix $\mathbf{Z}_P$ comprises of the vectors ζ_i^P and, $\mathbf{Z}_H$ comprises of

the vectors ζ_i^H . The matrix of vectors of the extended (adjoint) center subspace is denoted by, $\widehat{\mathbf{W}} = [\widehat{\mathbf{W}}_C, \widehat{\mathbf{W}}_P, \widehat{\mathbf{W}}_H]$ with the individual vectors denoted as $\widehat{\mathbf{w}}_i$, $i \in 1, 2, \dots, m$. It can be verified that,

$$\widehat{\mathbf{W}}^\dagger \widehat{\mathbf{V}} = \mathbf{I}_m. \quad (29)$$

This is the analogue of equation (6), and is simply a statement of the biorthogonality relation between the direct and adjoint eigenvectors in the extended center subspace.

Finally, we have the generalized eigenvalue relations,

$$\widehat{\mathbf{L}}\widehat{\mathbf{V}} = \widehat{\mathbf{V}}\widehat{\mathbf{K}}, \quad \widehat{\mathbf{L}}^\dagger \widehat{\mathbf{W}} = \widehat{\mathbf{W}}\widehat{\mathbf{K}}^\dagger \quad (30)$$

with,

$$\widehat{\mathbf{K}} = \widehat{\mathbf{W}}^\dagger \widehat{\mathbf{L}} \widehat{\mathbf{V}} = \begin{bmatrix} \Lambda_C & \Gamma_P & \Gamma_H \\ \mathbf{0} & \Lambda_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Lambda_H \end{bmatrix}, \quad (31)$$

being the reduced matrix representing the action of $\widehat{\mathbf{L}}$ in the extended center subspace. Obviously $\widehat{\mathbf{K}} \in \mathbb{C}^{m \times m}$ and, the diagonal elements of $\widehat{\mathbf{K}}$ represent the eigenvalues of the reduced system, that are the critical eigenvalues of the extended system.

Note that due to the special structure of the adjoint parameter and forcing modes, the projection of the extended nonlinearity $\widehat{\mathcal{N}}'$ onto the subspace spanned by the (direct) parameter and forcing modes vanishes identically. That is to say,

$$\widehat{\mathbf{W}}_P^\dagger \widehat{\mathcal{N}}' = \mathbf{0}, \quad \widehat{\mathbf{W}}_H^\dagger \widehat{\mathcal{N}}' = \mathbf{0}. \quad (32)$$

This completes the extension of the system and the subspace structure for both the direct and adjoint systems can be readily identified. The extension of the tangent space is summarized in algorithms 1 and 2.

While the structure of the extended subspaces has been evaluated, reformulation of the problem as an extended system may have detrimental practical implications when considering large systems. Practical methods for solving such systems often rely on numerical algorithms that exploit inherent structure of the problem (Hermicity for conjugate-gradient solvers for example). In such a scenario, additional degrees of freedom can potential destroy the system symmetries and may require

Algorithm 1 Extended tangent space ($\widehat{\mathbf{L}}$)

Require: System: $\partial \mathbf{u} / \partial t = \mathbf{L} \mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.

Ensure: $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $(\partial \mathcal{N} / \partial \mathbf{u})|_{\mathbf{u}=\mathbf{0}} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger \mathbf{L} \mathbf{V} = \Lambda_C$.

/ Center subspace dimension: n . */*

Construct: Linear operators $\mathbf{L}_\nu \in \mathbb{C}^{n \times p}$ and $\mathbf{L}_\theta \in \mathbb{C}^{n \times h}$ for extended variables ν, θ .

Require: Diagonal submatrices: Λ_P, Λ_H .

/ Build: $\Gamma_P \in \mathbb{C}^{n \times p}$; $\Gamma_H \in \mathbb{C}^{n \times h}$. */*

for ($i \in 1, \dots, p$) **do**

for ($j \in 1, \dots, n$) **do**

if ($\lambda_i^P = \lambda_j^C$) **then**

$(\Gamma_P)_{j,i} = \mathbf{w}_j^\dagger \mathbf{L}_\nu \mathbf{e}_i^P,$

else

$(\Gamma_P)_{j,i} = 0,$

end if

end for

end for

/ Solve $\mathbf{V}_P$ */*

$\mathbf{L} \mathbf{V}_P + \mathbf{L}_\nu \mathbf{I}_p = \mathbf{V}_P \Lambda_P + \mathbf{V} \Gamma_P,$ */* Eq. (17) */*

/ with singular components removed. */*

for ($i \in 1, \dots, h$) **do**

for ($j \in 1, \dots, n$) **do**

if ($\lambda_i^H = \lambda_j^C$) **then**

$(\Gamma_H)_{j,i} = (\gamma_i^H)_j = \mathbf{w}_j^\dagger \mathbf{L}_\theta \mathbf{e}_i^H,$

else

$(\Gamma_H)_{j,i} = (\gamma_i^H)_j = 0.$

end if

end for

end for

/ Solve $\mathbf{V}_H$ */*

$\mathbf{L} \mathbf{V}_H + \mathbf{L}_\theta \mathbf{I}_h = \mathbf{V}_H \Lambda_H + \mathbf{V} \Gamma_H,$ */* Eq. (20) */*

/ with singular components removed. */*

Construct: $\widehat{\mathbf{V}}$ */* Eq. (22) */*

Construct: $\widehat{\mathbf{K}}$ */* Eq. (31) */*

return $\widehat{\mathbf{K}}, \widehat{\mathbf{V}}$.

substantial changes in the numerical approach to the system solution. This would be particularly true of asymptotic approaches that require resolvent calculations [20, 24, 10, 7, 21, 34, 25, 26, 35], which itself can be a computationally challenging task for large systems. Such a scenario would be highly undesirable. However, as it turns out, at least some of the parameterizations involved in the evaluation of center-manifolds do not in fact

Algorithm 2 Extended tangent space ($\widehat{\mathbf{L}}^\dagger$)

Require: System: $\partial \mathbf{u} / \partial t = \mathbf{L} \mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.**Ensure:** $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $(\partial \mathcal{N} / \partial \mathbf{u})|_{\mathbf{u}=\mathbf{0}} = \mathbf{0}$.**Require:** Tangent Space: $\mathbf{W}^\dagger \mathbf{L} \mathbf{V} = \boldsymbol{\Lambda}_C$./* Center subspace dimension: n . */**Construct:** Linear operators $\mathbf{L}_\nu \in \mathbb{C}^{N \times p}$ and $\mathbf{L}_\theta \in \mathbb{C}^{N \times h}$ for extended variables ν, θ .**Require:** Diagonal submatrices: $\boldsymbol{\Lambda}_P, \boldsymbol{\Lambda}_H$./* Build: $\mathbf{Z}_P \in \mathbb{C}^{p \times n}$; $\mathbf{Z}_H \in \mathbb{C}^{h \times n}$. */

```
for ( $i \in 1, \dots, n$ ) do
  for ( $j \in 1, \dots, p$ ) do
    if ( $\lambda_j^P = \lambda_i^C$ ) then
      | ( $\mathbf{Z}_P$ ) $_{j,i} = 0$ ,
    else
      | ( $\mathbf{Z}_P$ ) $_{j,i} = (\mathbf{L}_\nu^\dagger \mathbf{w}_i)_j / (\lambda_i^{C*} - \lambda_j^{P*})$ ,
    end if
  end for
  for ( $j \in 1, \dots, h$ ) do
    if ( $\lambda_j^H = \lambda_i^C$ ) then
      | ( $\mathbf{Z}_H$ ) $_{j,i} = 0$ ,
    else
      | ( $\mathbf{Z}_H$ ) $_{j,i} = (\mathbf{L}_\theta^\dagger \mathbf{w}_i)_j / (\lambda_i^{C*} - \lambda_j^{H*})$ ,
    end if
  end for
end for
Construct:  $\widehat{\mathbf{W}}$  /* Eq. (28)
return  $\widehat{\mathbf{W}}$ .
```

require the use of the extended system after the initial evaluation of the extended tangent space. All higher order calculations can be performed using the original linear system $\mathbf{L}$. This is clarified in the next section where the parameterization method and its asymptotic terms are explicitly built.

4 Center Manifold for the Extended System

In the context of invariant manifold theory, several previous approaches for the approximation of such manifolds can be effectively viewed as instances of what is referred to as the parametrization method. The framework was introduced in a series of papers by Cabré et al. [22, 23, 24], although, a particular instance of such an approach was presented earlier by Coullet and Spiegel [20] for the evaluation of the normal form of center manifolds.

Nevertheless, subsequent works by Carini, Auteri, and Giannetti [21], Jain and Haller [26], Negi [35] can be fruitfully viewed through the lens of the parameterization method. Within this framework the reduced representation of the invariant manifold being approximated can be chosen in an arbitrary manner depending on the particular requirements of the reduction being sought. Amongst the plethora of arbitrary choices, the normal form is an important representation that is often sought since it results in the simplest form of the reduced equations. In the next subsection the normal form representation for the center-manifold of the extended system is deduced.

To simplify notation for multiple indices used hereafter, a multi-dimensional index set $\mathcal{I}\{(k), [a, b]\}$ is introduced, which is an *ordered* set of all k -tuples $(i_1, i_2, \dots, i_k)$, defined as,

$$\begin{aligned} \mathcal{I}\{(k), [a, b]\} := & \{(i_1, i_2, \dots, i_k) \mid a, b \in \mathbb{N}, \\ & i_1, i_2, \dots, i_k \in \mathbb{N}, \\ & a \leq i_i \leq b, i_1 \leq i_2 \leq b, \\ & \dots, i_{k-1} \leq i_k \leq b\}, \end{aligned} \quad (33)$$

where, $i_1, i_2, \dots, i_k$, are the index variables and $[a, b]$ specify the lower and upper bounds of the indices. Obviously the indices i_1 etc. are dummy indices however their relative position is important since the higher index always starts from the previous index. The set is ordered in the sense that the first element of the $\mathcal{I}\{(k), [a, b]\}$ is always $(i_1, i_2, \dots, i_k) = (a, a, \dots, a)$, the last element is $(i_1, i_2, \dots, i_k) = (b, b, \dots, b)$ and it is the last index (i_k) which iterates the fastest. Hence the index set $\mathcal{I}\{(3), [1, 3]\}$ is the set containing the tuples,

$$\begin{aligned} (i_1, i_2, i_3) \in & \{(1, 1, 1), (1, 1, 2), (1, 1, 3), \\ & (1, 2, 2), (1, 2, 3), (1, 3, 3), \\ & (2, 2, 2), (2, 2, 3), (2, 3, 3), \\ & (3, 3, 3)\} \end{aligned} ,$$

in that order. If an element $(i_1, i_2, \dots, i_k)$ precedes an element $(j_1, j_2, \dots, j_k)$ in ordering it is denoted as $(i_1, i_2, \dots, i_k) \prec (j_1, j_2, \dots, j_k)$ while, if $(i_1, i_2, \dots, i_k)$ succeeds $(j_1, j_2, \dots, j_k)$ in the ordering it is denoted as $(i_1, i_2, \dots, i_k) \succ (j_1, j_2, \dots, j_k)$.

In a similar vein, multiple nested summations are denoted as

$$\sum_{(k)}^{[a,b]} = \sum_{i_1=a}^b \sum_{i_2=i_1}^b \cdots \sum_{i_k=i_{k-1}}^b \quad (34)$$

As before, the higher index starts from the previous index. In principle the ordering of the tuples in a particular index set does not matter to the overall results however, keeping the ordering in mind helps to elucidate the structure of the asymptotic problem as will be shown subsequently.

For the asymptotic expansions we will typically have $a = 1$ and $b = m$, where m is the dimension of the manifold. For such cases the number of k -tuples in $\mathcal{I}\{(k), [1, m]\}$ is analytically known to be $T_m^k = C(m + k - 1, k)$, where,

$$C(p, q) = \binom{p}{q} = \frac{p!}{q!(p-q)!}, \quad (35)$$

defines the binomial coefficient of p and q .

4.1 Normal Form

One may find the exposition on the implementation of the parameterization method in previous works of Haro et al. [25] and Jain and Haller [26]. To be clear, the application of the parameterization method to an extended system like equation (10) follows the same steps as outlined in those earlier works. However, many of the essential steps are repeated here to bring out the structure of system of equations at each asymptotic order in the context of system extension.

Consider the extended system as defined by equation (10), represented in a compact form by equation (12), which has an m -dimensional critical subspace ($m = n + p + h$). Assume $\mathbf{z} = [z_1; z_2; \dots; z_m]$ is an m -dimensional vector of coordinates that parameterize the center-manifold of the extended system. The evolution of the system along the center-manifold in the parameterized coordinates is assumed to be given by

$$\begin{aligned} \dot{\mathbf{z}} &= \mathcal{G}(\mathbf{z}), \\ \mathcal{G} : \mathbb{C}^m &\rightarrow \mathbb{C}^m. \end{aligned} \quad (36)$$

$\mathcal{G}$ will also be referred to as the reduced representation (of the extended system). The system

solution on the center-manifold corresponding to a point $\mathbf{z}$ of the parameterized system is given by $\mathcal{Y}(\mathbf{z})$, which is map from the parameterized coordinates to the full system state,

$$\mathcal{Y} : \mathbb{C}^m \rightarrow \mathbb{C}^M. \quad (37)$$

At this stage neither $\mathcal{G}(\mathbf{z})$ nor $\mathcal{Y}(\mathbf{z})$ have been specified in any form, since, the parameterization can be arbitrarily chosen. In order to obtain a normal form of the reduced system one may expand both $\mathcal{G}(\mathbf{z})$ and $\mathcal{Y}(\mathbf{z})$ as a formal power series in $\mathbf{z}$ and constrain the asymptotic form of $\mathcal{G}$ to be in a normal form for the parameterized coordinates $\mathbf{z}$. The asymptotic form of $\mathcal{Y}$ is obtained by introducing the asymptotic expansions into equation (12) and solving the terms order-by-order.

In what follows, Einstein summation is *not* assumed for repeated indices unless otherwise specified. We assume the power series expansions as,

$$\begin{aligned} \mathcal{G}(\mathbf{z}) &= \sum_{(1)}^{[1,m]} \mathbf{g}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \mathbf{g}_{i_1, i_2} z_{i_1} z_{i_2} + \\ &\sum_{(3)}^{[1,m]} \mathbf{g}_{i_1, i_2, i_3} z_{i_1} z_{i_2} z_{i_3} + \dots, \end{aligned} \quad (38)$$

and,

$$\begin{aligned} \mathcal{Y}(\mathbf{z}) &= \sum_{(1)}^{[1,m]} \hat{\mathbf{y}}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \hat{\mathbf{y}}_{i_1, i_2} z_{i_1} z_{i_2} + \\ &\sum_{(3)}^{[1,m]} \hat{\mathbf{y}}_{i_1, i_2, i_3} z_{i_1} z_{i_2} z_{i_3} + \dots \end{aligned} \quad (39)$$

Here the terms $\mathbf{g}_{i_1, i_2, i_3, \dots} \in \mathbb{C}^m$, while, $\hat{\mathbf{y}}_{i_1, i_2, i_3, \dots} \in \mathbb{C}^M$. Given the asymptotic form of $\mathcal{Y}(\mathbf{z})$, one automatically obtains the nonlinear term $\widehat{\mathcal{N}}'$ as a power series in $\mathbf{z}$. This can be represented as,

$$\begin{aligned} \widehat{\mathcal{N}}' &= \sum_{(2)}^{[1,m]} \hat{\mathbf{h}}_{i_1, i_2} z_{i_1} z_{i_2} + \\ &\sum_{(3)}^{[1,m]} \hat{\mathbf{h}}_{i_1, i_2, i_3} z_{i_1} z_{i_2} z_{i_3} + \dots, \end{aligned} \quad (40)$$

which is obtained via the collection of terms with similar powers.

Substitution of the power series expansions into equation (12) results in

$$\sum_{\alpha=1}^m \frac{\partial}{\partial z_{\alpha}} \left(\sum_{(1)}^{[1,m]} \hat{\mathbf{y}}_{i_1} z_{i_1} + \dots \right) \left(\sum_{(1)}^{[1,m]} g_{j_1}^{\alpha} z_{j_1} + \dots \right) = \hat{\mathbf{L}} \left(\sum_{(1)}^{[1,m]} \hat{\mathbf{y}}_{i_1} z_{i_1} + \dots \right) + \left(\sum_{(2)}^{[1,m]} \hat{\mathbf{h}}_{i_1, i_2} z_{i_1} z_{i_2} + \dots \right). \quad (41)$$

Here $g_{j_1}^{\alpha}$ represents the α^{th} element of the vector $\mathbf{g}_{j_1}$. At order k , one obtains $m \times T_m^k$ unknowns which define the elements of the vectors $\mathbf{g}_{i_1, \dots, i_k}$, and the corresponding T_m^k unknown vectors $\hat{\mathbf{y}}_{i_1, \dots, i_k}$. The definitions of the two sets of unknowns are interdependent since the form of the reduced representation $\mathcal{G}(\mathbf{z})$ determines the map $\mathcal{Y}(\mathbf{z})$ from the parameterized coordinates to the full solution (and vice versa).

4.1.1 Asymptotic terms of $\mathcal{O}(|z|)$

At first order the non-linear terms do not contribute and one simply obtains,

$$\sum_{(1)}^{[1,m]} \left(\sum_{\alpha=1}^m g_{i_1}^{\alpha} \hat{\mathbf{y}}_{\alpha} - \hat{\mathbf{L}} \hat{\mathbf{y}}_{i_1} \right) z_{i_1} = \mathbf{0}. \quad (42)$$

Here, the first restrictions on the asymptotic form of $\mathcal{G}$ are applied, associating the unknown terms $\hat{\mathbf{y}}_{i_1}$, $\mathbf{g}_{i_1}$ etc. with the generalized eigenvalue problem such that,

$$[\hat{\mathbf{y}}_{i_1}, \hat{\mathbf{y}}_{i_2}, \dots, \hat{\mathbf{y}}_{i_m}] = \hat{\mathbf{V}}, \quad (43a)$$

$$[\mathbf{g}_{i_1}, \mathbf{g}_{i_2}, \dots, \mathbf{g}_{i_m}] = \hat{\mathbf{K}}. \quad (43b)$$

This reduces equation (42) to equation (30) which, then satisfies the tangency condition of the invariant manifold, *i.e.*, at the origin the center-manifold is tangent to the center subspace. Subsequently, $\hat{K}_{i,j}$ will be used for all the first order terms of the type $\mathbf{g}_j^i$ and the first order vectors $\hat{\mathbf{y}}_i$ will be replaced by the respective eigenvectors $\hat{\mathbf{v}}_i$. The reduced matrix $\hat{\mathbf{K}}$ is responsible for the cross-coupling of different $\hat{\mathbf{y}}_{i_1, \dots, i_k}$ at each order of the asymptotic expansion.

4.1.2 Asymptotic terms of $\mathcal{O}(|z|^2)$

At second order the non-linear terms contribute and one obtains the following equation,

$$\begin{aligned} & \sum_{(2)}^{[1,m]} \sum_{j=1}^m \left(\hat{K}_{i_1, j} z_{i_2} z_j + \hat{K}_{i_2, j} z_{i_1} z_j \right) \hat{\mathbf{y}}_{i_1, i_2} + \\ & \sum_{(2)}^{[1,m]} \sum_{j=1}^m g_{i_1, i_2}^j \hat{\mathbf{v}}_j z_{i_1} z_{i_2} - \sum_{(2)}^{[1,m]} \hat{\mathbf{L}} \hat{\mathbf{y}}_{i_1, i_2} z_{i_1} z_{i_2} \quad (44) \\ & = \sum_{(2)}^{[1,m]} \hat{\mathbf{h}}_{i_1, i_2} z_{i_1} z_{i_2}. \end{aligned}$$

Collecting the expressions for a particular asymptotic term $z_a z_b$, where $(a, b) \in \mathcal{I}\{(2), [1, m]\}$, equation (44) reduces to,

$$\begin{aligned} & \left[\sum_{i=1}^a \hat{K}_{i, b} \hat{\mathbf{y}}_{i, a} + \sum_{i=1}^b \hat{K}_{i, a} \hat{\mathbf{y}}_{i, b} - \sum_{i=1}^a \hat{K}_{i, b} \hat{\mathbf{y}}_{i, a} \delta_b^a \right] + \\ & \left[\sum_{i=a}^m \hat{K}_{i, b} \hat{\mathbf{y}}_{a, i} + \sum_{i=b}^m \hat{K}_{i, a} \hat{\mathbf{y}}_{b, i} - \sum_{i=b}^m \hat{K}_{i, a} \hat{\mathbf{y}}_{b, i} \delta_b^a \right] \quad (45) \\ & + \sum_{j=1}^m g_{a, b}^j \hat{\mathbf{v}}_j - \hat{\mathbf{L}} \hat{\mathbf{y}}_{a, b} = \hat{\mathbf{h}}_{a, b}. \end{aligned}$$

At second order one obtains T_m^2 equations, one for each tuple $(a, b) \in \mathcal{I}\{(2), [1, m]\}$, which collectively form the system of equations for all $\hat{\mathbf{y}}_{i_1, i_2}$ vectors that needs to be solved. Here, several properties of the structure of equation (45) need to be pointed out that have important practical implications.

First, the various unknown vectors $\hat{\mathbf{y}}_{i_1, i_2}$ are coupled due to the off-diagonal terms in $\hat{\mathbf{K}}$, which appear in the first two bracketted terms in equation (45). Recall that we are considering the tuples in our index set to be ordered so that the vectors $\hat{\mathbf{y}}_{i_1, i_2}$ can also be ordered in the same manner. For any index $(a, b) \in \mathcal{I}\{(2), [1, m]\}$ (and assuming $b > a$) the bracketted terms in

equation (45) can be rewritten as,

$$\begin{aligned}
& \underbrace{\sum_{i=1}^a \left(\hat{K}_{i,b} \hat{\mathbf{y}}_{i,a} - \hat{K}_{i,b} \hat{\mathbf{y}}_{i,a} \delta_b^a \right)}_{P1} + \underbrace{\sum_{i=1}^{a-1} \hat{K}_{i,a} \hat{\mathbf{y}}_{i,b}}_{P2} + \\
& \underbrace{\hat{K}_{a,a} \hat{\mathbf{y}}_{a,b}}_{D1} + \underbrace{\sum_{i=a+1}^b \hat{K}_{i,a} \hat{\mathbf{y}}_{i,b}}_{S1} + \\
& \underbrace{\sum_{i=a}^{b-1} \hat{K}_{i,b} \hat{\mathbf{y}}_{a,i}}_{P3} + \underbrace{\hat{K}_{b,b} \hat{\mathbf{y}}_{a,b}}_{D2} + \underbrace{\sum_{i=b+1}^m \hat{K}_{i,b} \hat{\mathbf{y}}_{a,i}}_{S2} + \\
& \underbrace{\sum_{i=b}^m \left(\hat{K}_{i,a} \hat{\mathbf{y}}_{b,i} - \hat{K}_{i,a} \hat{\mathbf{y}}_{b,i} \delta_b^a \right)}_{S3}.
\end{aligned} \tag{46}$$

The terms marked as $P1, P2, P3$ are terms where elements of $\hat{\mathbf{K}}$ multiply vectors $\hat{\mathbf{y}}_{i_1, i_2}$ such that $(i_1, i_2) \prec (a, b)$. The terms marked $D1, D2$ are those where elements of $\hat{\mathbf{K}}$ multiply only with $\hat{\mathbf{y}}_{a,b}$, and the terms marked as $S1, S2, S3$ are those where $\hat{\mathbf{K}}$ elements multiply vectors succeeding $\hat{\mathbf{y}}_{a,b}$, *i.e.*, $(i_1, i_2) \succ (a, b)$. Notice that all the succeeding terms in $S1, S2, S3$ are always multiplied by elements of the $\hat{\mathbf{K}}$ matrix that are below the main diagonal. However, since $\hat{\mathbf{K}}$ has an upper triangular structure, all terms marked $S1, S2, S3$ vanish. Therefore the equation resulting from any polynomial term $z_a z_b$ couples the vector $\hat{\mathbf{y}}_{a,b}$ only with those that precede it in the ordering, *i.e.*, with $\hat{\mathbf{y}}_{i_1, i_2} \prec \hat{\mathbf{y}}_{a,b}$. Thus the resulting system of equations for the T_m^2 unknown vectors $\hat{\mathbf{y}}_{i_1, i_2}$ at second order is lower triangular and even though the fields are coupled with each other, they can still be solved for sequentially one at a time with back substitution. The splitting of terms in equation (46) is done for $b > a$ however, the same conclusion applies even for the case of $b = a$ and the resulting system of equations is always lower triangular.

A visual representation of this cross coupling is shown in figure 1. For a given index $(a, b) \in \mathcal{I}\{(2), [1, m]\}$, the elements of the matrix $\hat{\mathbf{K}}$ marked by $\mathbf{P}$ multiply vectors $\hat{\mathbf{y}}_{i_1, i_2}$ that precede $\hat{\mathbf{y}}_{a,b}$ in the ordering. Similarly, the elements marked by $\mathbf{S}$ multiply the vectors $\hat{\mathbf{y}}_{i_1, i_2}$ which

Fig. 1: Coupling of the vectors $\hat{\mathbf{y}}_{i_1, i_2}$ with the $\hat{\mathbf{K}}$ matrix for a given (a, b) .

succeed $\hat{\mathbf{y}}_{a,b}$ in the ordering. The diagonal terms $\hat{K}_{a,a}$ and $\hat{K}_{b,b}$ always multiply $\hat{\mathbf{y}}_{a,b}$. Obviously, when $\hat{\mathbf{K}}$ is upper triangular, the contributions of the succeeding elements will always vanish. This lower triangular structure of the resulting system of equations for $\hat{\mathbf{y}}_{i_1, i_2}$ is not confined to the second-order approximation but in fact persists even for higher order asymptotic approximations so that at all orders, the system of equations can always be solved sequentially, one vector at a time. This is an important aspect particularly for large systems since the computational cost of a simultaneous coupled system solve can be prohibitive (as T_m^k becomes large).

To keep the notation compact, the non-vanishing off-diagonal coupling terms will be represented as $\hat{\mathbf{h}}_{a,b}^O$, defined for second order as,

$$\hat{\mathbf{h}}_{a,b}^O = \begin{cases} \sum_{i=1}^a \left(\hat{K}_{i,b} \hat{\mathbf{y}}_{i,a} - \hat{K}_{i,b} \hat{\mathbf{y}}_{i,a} \delta_b^a \right) \\ + \sum_{i=1}^{a-1} \hat{K}_{i,a} \hat{\mathbf{y}}_{i,b} + \sum_{i=a}^{b-1} \hat{K}_{i,b} \hat{\mathbf{y}}_{a,i}. \end{cases} \tag{47}$$

The resulting equation for $\hat{\mathbf{y}}_{a,b}$ may be written as

$$[\hat{\mathcal{R}}(\omega_{a,b})]^{-1} \hat{\mathbf{y}}_{a,b} + \sum_{i=1}^m g_{a,b}^i \hat{\mathbf{v}}_i = \hat{\mathbf{h}}_{a,b} - \hat{\mathbf{h}}_{a,b}^O, \tag{48a}$$

$$\omega_{a,b} = \hat{K}_{a,a} + \hat{K}_{b,b}, \tag{48b}$$

where, $\widehat{\mathcal{R}}(\omega)$ is the resolvent operator for the extended linear system defined as,

$$\widehat{\mathcal{R}}(\omega) = (\omega \widehat{\mathbf{I}} - \widehat{\mathbf{L}})^{-1}. \quad (49)$$

The resolvent operator is the key mathematical object defining the asymptotic expressions. $\widehat{\mathcal{R}}(\omega_{a,b})$ is singular when $\omega_{a,b}$ is an eigenvalue of $\widehat{\mathbf{L}}$, which is often referred to as a resonance condition. The normal form of the invariant manifold precisely resolves this singularity by requiring $\widehat{\mathbf{y}}_{a,b}$ to have a vanishing component along the singular direction. However, equation (45) still needs to be balanced along the singular directions, which can only be done by $\sum_{j=1}^m g_{a,b}^j \widehat{\mathbf{v}}_j$. Multiplying equation (48) with $\widehat{\mathbf{w}}_j^\dagger$ and requiring that $\widehat{\mathbf{w}}_j^\dagger \widehat{\mathbf{y}}_{a,b} = 0$ (in case of resonance), then provides us with the definition of $\mathbf{g}_{a,b}$,

$$g_{a,b}^j = \begin{cases} \widehat{\mathbf{w}}_j^\dagger (\widehat{\mathbf{h}}_{a,b} - \widehat{\mathbf{h}}_{a,b}^O); & \text{if } \omega_{a,b} = \widehat{K}_{j,j}, \\ 0, & \text{otherwise,} \end{cases} \quad (50)$$

where, $\widehat{\mathbf{w}}_j$ is the j^{th} adjoint eigenvector of $\widehat{\mathbf{L}}$ as defined earlier. Since the system is lower triangular the terms $\widehat{\mathbf{h}}_{a,b}^O$ are always known apriori when solving the system sequentially.

4.1.3 Asymptotic terms of $\mathcal{O}(|z|^3)$

At third (and higher) order one also obtains contributions from the lower order terms. The sum of all contributions at third order can be written as,

$$\begin{aligned} & \sum_{(3)}^{[1,m]} \sum_{j=1}^m \left[\widehat{K}_{i_1,j} z_{i_2} z_{i_3} z_j + \widehat{K}_{i_2,j} z_{i_1} z_{i_3} z_j + \right. \\ & \left. \widehat{K}_{i_3,j} z_{i_1} z_{i_2} z_j \right] \widehat{\mathbf{y}}_{i_1,i_2,i_3} + \\ & \sum_{(2)}^{[1,m]} \sum_{(2)}^{[1,m]} \left[g_{j_1,j_2}^{i_1} z_{i_2} z_{j_1} z_{j_2} + g_{j_1,j_2}^{i_2} z_{i_1} z_{j_1} z_{j_2} \right] \widehat{\mathbf{y}}_{i_1,i_2} \\ & + \sum_{j=1}^m \sum_{(3)}^{[1,m]} g_{i_1,i_2,i_3}^j \widehat{\mathbf{v}}_j z_{i_1} z_{i_2} z_{i_3} \\ & - \sum_{(3)}^{[1,m]} \widehat{\mathbf{L}} \widehat{\mathbf{y}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} = \sum_{(3)}^{[1,m]} \widehat{\mathbf{h}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3}. \end{aligned} \quad (51)$$

Fig. 2: Coupling of the vectors $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ with the $\widehat{\mathbf{K}}$ matrix for a given (a,b,c) .

The $\widehat{\mathbf{K}}$ matrix is responsible for the cross contributions of different vectors $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ as seen from the first set of bracketted terms in equation (51). Again, for a given tuple $(a,b,c) \in \mathcal{I}\{(3), [1,m]\}$, the coupling of $\widehat{\mathbf{K}}$ elements with vectors preceding and succeeding $\widehat{\mathbf{y}}_{a,b,c}$ is depicted in figure 2. With $\widehat{\mathbf{K}}$ being upper triangular, the contribution of succeeding elements, $(i_1, i_2, i_3) \succ (a, b, c)$, vanishes and the resulting system of equations for $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ at third order is again lower triangular.

The second set of bracketted terms in equation (51) are contributions from the lower order asymptotic terms. For a given index (a,b,c) , we represent the low order contributions as $\widehat{\mathbf{h}}_{a,b,c}^L$ and the contributions of the off-diagonal terms as $\widehat{\mathbf{h}}_{a,b,c}^O$ resulting in the compact expression for $\widehat{\mathbf{y}}_{a,b,c}$ as

$$[\widehat{\mathcal{R}}(\omega_{a,b,c})]^{-1} \widehat{\mathbf{y}}_{a,b,c} + \sum_{j=1}^m g_{a,b,c}^j \widehat{\mathbf{v}}_j \quad (52a)$$

$$\begin{aligned} &= \widehat{\mathbf{h}}_{a,b,c} - \widehat{\mathbf{h}}_{a,b,c}^O - \widehat{\mathbf{h}}_{a,b,c}^L, \\ \omega_{a,b,c} &= \widehat{K}_{a,a} + \widehat{K}_{b,b} + \widehat{K}_{c,c} \end{aligned} \quad (52b)$$

The resolution of the system singularities due to the resonance condition, $\omega_{a,b,c} = \widehat{K}_{j,j}$, provides us with the third order terms of the reduced

representation,

$$g_{a,b,c}^j = \begin{cases} \widehat{\mathbf{w}}_j^\dagger (\widehat{\mathbf{h}}_{a,b,c} - \widehat{\mathbf{h}}_{a,b,c}^L - \widehat{\mathbf{h}}_{a,b,c}^O); & \text{if } \omega_{a,b,c} = \widehat{K}_{j,j} \\ 0, & \text{otherwise,} \end{cases} \quad (53)$$

and $\widehat{\mathbf{w}}_j^\dagger \widehat{\mathbf{y}}_{a,b,c} = 0$ if the resonance condition holds.

4.1.4 Asymptotic terms of $\mathcal{O}(|z|^k)$

For all subsequent orders the structure of the problem remains essentially the same as the third order terms. At any order k , one obtains a systems of equations for T_m^k vectors $\widehat{\mathbf{y}}_{i_1, \dots, i_k}$ which is lower triangular and hence can be solved sequentially. The cross contributions of the vectors due to $\widehat{\mathbf{K}}$ has the same structure as in figure 2 but with k columns (possibly with repetitions) of $\widehat{\mathbf{K}}$ contributing. The lower order terms $\widehat{\mathbf{h}}_{i_1, \dots, i_k}^L$ are always known a-priori. When solving for the unknown vectors sequentially the off-diagonal contributions $\widehat{\mathbf{h}}_{i_1, \dots, i_k}^O$ are also known a-priori. For each index $(i_1, \dots, i_k) \in \mathcal{I}\{(k), [1, m]\}$ one obtains an equation similar to equation (52),

$$[\widehat{\mathcal{R}}(\omega_{i_1, \dots, i_k})]^{-1} \widehat{\mathbf{y}}_{i_1, \dots, i_k} + \sum_{j=1}^m g_{i_1, \dots, i_k}^j \widehat{\mathbf{v}}_j \quad (54a)$$

$$= \widehat{\mathbf{h}}_{i_1, \dots, i_k} - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^O - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^L,$$

$$\omega_{i_1, \dots, i_k} = \widehat{K}_{i_1, i_1} + \widehat{K}_{i_2, i_2} + \dots + \widehat{K}_{i_k, i_k}, \quad (54b)$$

with the resonance condition arising whenever,

$$\omega_{i_1, \dots, i_k} = \widehat{K}_{j,j}, \quad j \in 1, \dots, m, \quad (55)$$

which then defines the normal form coefficients,

$$g_{i_1, \dots, i_k}^j = \begin{cases} \widehat{\mathbf{w}}_j^\dagger (\widehat{\mathbf{h}}_{i_1, \dots, i_k} - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^L - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^O); & \text{if } \omega_{i_1, \dots, i_k} = \widehat{K}_{j,j} \\ 0 = & \text{otherwise,} \end{cases} \quad (56)$$

and, $\widehat{\mathbf{w}}_j^\dagger \widehat{\mathbf{y}}_{i_1, \dots, i_k} = 0$ in case of resonance.

4.2 From the Extended System to the Standard System

For all orders $k \geq 2$, one needs to evaluate a resolvent equation of the type given by equation (54) for a given right hand side. The $\widehat{\cdot}$ symbol on

all terms indicates that the resolvent needs to be evaluated for the extended system. This can potentially be a major drawback since numerical methods for large systems often rely on exploiting the inherent symmetry of the operators which may be destroyed by system extension. However, as it turns out, even though the resolvents are technically solved for the extended linear system $\widehat{\mathbf{L}}$, in fact only the resolvent solutions of the original linear operator $\mathbf{L}$ are required. This can be shown by examining the terms sequentially.

First we express equation (54) in its three component form for a generic vector $\widehat{\mathbf{y}} = [\mathbf{y}^{(1)}; \mathbf{y}^{(2)}; \mathbf{y}^{(3)}]$, for a particular frequency ω and generic right hand side, $\widehat{\mathbf{r}} = [\mathbf{r}^{(1)}; \mathbf{r}^{(2)}; \mathbf{r}^{(3)}]$ as,

$$(\omega \mathbf{I} - \mathbf{L}) \mathbf{y}^{(1)} + \mathbf{L}_\nu \mathbf{y}^{(2)} + \mathbf{L}_\theta \mathbf{y}^{(3)} = \mathbf{r}^{(1)}, \quad (57a)$$

$$(\omega \mathbf{I} - \mathbf{A}_P) \mathbf{y}^{(2)} = \mathbf{r}^{(2)}, \quad (57b)$$

$$(\omega \mathbf{I} - \mathbf{A}_H) \mathbf{y}^{(3)} = \mathbf{r}^{(3)}. \quad (57c)$$

Clearly, $\mathbf{r}^{(2)} = \mathbf{0}$ and an ω that is non-resonant with λ^P implies $\mathbf{y}^{(2)} = \mathbf{0}$. Similarly, $\mathbf{r}^{(3)} = \mathbf{0}$ implies $\mathbf{y}^{(3)} = \mathbf{0}$ when ω is non-resonant with the external forcing frequencies λ^H . In either case, if ω is resonant with the frequencies in λ^P or λ^H , the condition for the removal of singularities, $\widehat{\mathbf{w}}_j^\dagger \widehat{\mathbf{y}} = \mathbf{0}$, again ensures $\mathbf{y}^{(2)} = \mathbf{0}$ and $\mathbf{y}^{(3)} = \mathbf{0}$. Therefore equation (57) simply reduces to,

$$[\mathcal{R}(\omega)]^{-1} \mathbf{y}^{(1)} = \mathbf{r}^{(1)}; \quad \mathcal{R}(\omega) = (\omega \mathbf{I} - \mathbf{L})^{-1}, \quad (58)$$

where, $\mathcal{R}$ is now the resolvent operator for the standard linear system $\mathbf{L}$.

The asymptotic terms can now be examined sequentially noting that by construction the extended variables vanish for the non-linear term, which has the form $\widehat{\mathcal{N}}' = [\mathcal{N}'; \mathbf{0}; \mathbf{0}]$, and hence so do all $\widehat{\mathbf{h}}_{i_1, \dots, i_k} = [\mathbf{h}_{i_1, \dots, i_k}; \mathbf{0}; \mathbf{0}]$ derived from it. Starting at $\mathcal{O}(|z|^2)$ and $(i_1, i_2) = (1, 1)$, one has the resolvent equation in the extended system,

$$[\widehat{\mathcal{R}}(\omega_{1,1})]^{-1} \widehat{\mathbf{y}}_{1,1} + \sum_{j=1}^m g_{1,1}^j \widehat{\mathbf{v}}_j = \widehat{\mathbf{h}}_{1,1} \quad (59)$$

The extended variables vanish for $\widehat{\mathbf{h}}_{1,1}$ and therefore, regardless of resonance, they also vanish for $\widehat{\mathbf{y}}_{1,1}$, which then has the form $\widehat{\mathbf{y}}_{1,1} = [\mathbf{y}_{1,1}; \mathbf{0}; \mathbf{0}]$.

Hence equation (59) reduces to,

$$[\mathcal{R}(\omega_{1,1})]^{-1} \mathbf{y}_{1,1} + \sum_{j=1}^n g_{1,1}^j \mathbf{v}_j = \mathbf{h}_{1,1}, \quad (60)$$

with $g_{1,1}^j = 0$, for $j \in \{n+1, \dots, m\}$.

For the next term $(i_1, i_2) = (1, 2)$, the right hand side now also contains contributions $\hat{\mathbf{h}}_{1,2}^O$ due to the off-diagonal terms of $\hat{\mathbf{K}}$. These however can only come from the preceding term, $\hat{\mathbf{y}}_{1,1}$, and therefore off-diagonal contributions also have the form $\hat{\mathbf{h}}_{1,2}^O = [\mathbf{h}_{1,2}^O; \mathbf{0}; \mathbf{0}]$, which further implies that $\hat{\mathbf{y}}_{1,2} = [\mathbf{y}_{1,2}; \mathbf{0}; \mathbf{0}]$. The resulting equation being,

$$[\mathcal{R}(\omega_{1,2})]^{-1} \mathbf{y}_{1,2} + \sum_{j=1}^n g_{1,2}^j \mathbf{v}_j = \mathbf{h}_{1,2} - \mathbf{h}_{1,2}^O, \quad (61)$$

with $g_{1,2}^j = 0$, for $j \in \{n+1, \dots, m\}$.

This logic repeats for all $(i_1, i_2) \in \mathcal{I}\{(2), [1, m]\}$ and all unknown vectors at second order have the form $\hat{\mathbf{y}}_{i_1, i_2} = [\mathbf{y}_{i_1, i_2}; \mathbf{0}; \mathbf{0}]$ and therefore the resolvent equations can be solved using the original linear system $\mathbf{L}$.

At $\mathcal{O}(|\mathbf{z}|^3)$ the right hand side also has contributions from $\hat{\mathbf{h}}_{i_1, i_2, i_3}^L$ that arise from the lower order asymptotic terms. However, these arise only due to the second order asymptotic terms $\hat{\mathbf{y}}_{i_1, i_2}$ and therefore, $\hat{\mathbf{h}}_{i_1, i_2, i_3}^L = [\mathbf{h}_{i_1, i_2, i_3}^L; \mathbf{0}; \mathbf{0}]$. The same logical arguments presented earlier apply and hence all third order asymptotic terms also have the form $\hat{\mathbf{y}}_{i_1, i_2, i_3} = [\mathbf{y}_{i_1, i_2, i_3}; \mathbf{0}; \mathbf{0}]$ and only require the standard linear system $\mathbf{L}$ for their evaluation. As noted earlier, equations for the higher order asymptotic terms have the same essential structure as the third-order equations and hence at all orders the asymptotic terms have the form $\hat{\mathbf{y}}_{i_1, \dots, i_k} = [\mathbf{y}_{i_1, \dots, i_k}; \mathbf{0}; \mathbf{0}]$ and only require the standard linear system $\mathbf{L}$ for their evaluation.

Equations (54) and (56) may be rewritten for the standard system as,

$$[\mathcal{R}(\omega_{i_1, \dots, i_k})]^{-1} \mathbf{y}_{i_1, \dots, i_k} + \sum_{j=1}^n g_{i_1, \dots, i_k}^j \mathbf{v}_j \quad (62a)$$

$$= \mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^O - \mathbf{h}_{i_1, \dots, i_k}^L, \quad (62b)$$

$$\omega_{i_1, \dots, i_k} = \hat{K}_{i_1, i_1} + \hat{K}_{i_2, i_2} + \dots + \hat{K}_{i_k, i_k}.$$

with the resonance condition being the same, *i.e.*,

$$\omega_{i_1, \dots, i_k} = \hat{K}_{j, j}, \quad j \in 1, \dots, m, \quad (63)$$

which then defines the normal form coefficients for $j \in \{1, \dots, n\}$,

$$g_{i_1, \dots, i_k}^j = \begin{cases} \mathbf{w}_j^\dagger (\mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^L - \mathbf{h}_{i_1, \dots, i_k}^O) & \text{if } \omega_{i_1, \dots, i_k} = \hat{K}_{j, j} \\ 0 & \text{otherwise,} \end{cases} \quad (64)$$

and $\mathbf{w}_j^\dagger \mathbf{y}_{i_1, \dots, i_k} = 0$ in case of resonance. For all $j \in \{n+1, \dots, m\}$,

$$g_{i_1, \dots, i_k}^j = 0, \quad (65)$$

regardless of resonance.

The entire procedure for the evaluation of the normal form is presented succinctly in algorithm 3.

The fact that the extended variables vanish for all asymptotic vectors $\hat{\mathbf{y}}_{i_1, \dots, i_k}$ with $k \geq 2$ has another simplifying consequence for the parameter perturbations $\boldsymbol{\nu}$ and harmonic forcing amplitudes $\boldsymbol{\theta}$. Since these variables have been included into the state vector $\hat{\mathbf{u}}$, asymptotic evaluation of the state vector means that these extended variables are also evaluated asymptotically. Therefore,

$$\nu_j = \sum_{(1)}^m (\hat{\mathbf{y}}_{i_1})_{N+j} z_{i_1} + \sum_{(2)}^m (\hat{\mathbf{y}}_{i_1, i_2})_{N+j} z_{i_1} z_{i_2} + \dots, \quad (66)$$

where, $(\hat{\mathbf{y}}_{i_1, i_2})_{N+j}$ represents the $(N+j)^{th}$ component of the vector $\hat{\mathbf{y}}_{i_1, i_2}$. The high order vectors $\hat{\mathbf{y}}_{i_1, \dots, i_k}$ with $k \geq 2$ do not contribute while, for the linear term, ($k = 1$), only one eigenvector has a non-vanishing contribution to ν_j . Therefore we have a simple relation between the parameter perturbations ν_j and the reduced coordinates $\mathbf{z}$ as,

$$\nu_j = z_{n+j}. \quad (67)$$

In an identical manner, the relation between the forcing mode variables θ_j and the reduced coordinates is simply,

$$\theta_j = z_{n+p+j}. \quad (68)$$

Algorithm 3 Center–Manifold: Normal form

Require: System: $\partial \mathbf{u} / \partial t = \mathbf{L} \mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.

Ensure: $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $\partial \mathcal{N} / \partial \mathbf{u} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger \mathbf{L} \mathbf{V} = \boldsymbol{\Lambda}_C$.

/ Center subspace dimension: n . */*

Require: Extension via $\boldsymbol{\nu}, \boldsymbol{\theta}$ and $\mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta})$.

Evaluate $\widehat{\mathbf{V}}, \widehat{\mathbf{K}}$ via algorithm 1.

/ Center subspace dim.: $m = n + p + h$. */*

Require: Maximum Asymptotic order: k_{max}

/ Substitute asymptotic expansions */*

for $(k \in 2, 3, \dots, k_{max})$ **do**

for $(i_1, \dots, i_k) \in \mathcal{I}\{(k), [1, m]\}$ **do**

/ Build all r.h.s. contributions. */*

$\mathbf{r} = \mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^O - \mathbf{h}_{i_1, \dots, i_k}^L$

$\omega = \widehat{K}_{i_1, i_1} + \widehat{K}_{i_2, i_2} + \dots + \widehat{K}_{i_k, i_k}$

/ Build $\mathbf{g}_{i_1, \dots, i_k}$, Eq. (64) */*

for $(j \in 1, \dots, n)$ **do**

if $(\omega = \widehat{K}_{j, j})$ **then**

$g_{i_1, \dots, i_k}^j = \mathbf{w}_j^\dagger \mathbf{r}$

else

$g_{i_1, \dots, i_k}^j = 0$

end if

end for

for $(j \in n + 1, \dots, m)$ **do**

$g_{i_1, \dots, i_k}^j = 0$

end for

/ Build singular components */*

$\mathbf{r}_s = \sum_{j=1}^n \mathbf{v}_j g_{i_1, \dots, i_k}^j$

/ Evaluate resolvent */*

$\mathbf{y}_{i_1, \dots, i_k} = \mathcal{R}(\omega)[\mathbf{r} - \mathbf{r}_s]$ */* Eq. (62) */*

/ $\mathcal{R}(\omega) = (\omega \mathbf{I} - \mathbf{L})^{-1}$ */*

$\widehat{\mathbf{y}}_{i_1, \dots, i_k} = [\mathbf{y}_{i_1, \dots, i_k}; \mathbf{0}; \mathbf{0}]$

end for

end for

return $\mathcal{G}(\mathbf{z}), \mathcal{Y}(\mathbf{z})$.

4.3 A Graph Style Parameterization

In the earlier subsection it was shown that even though one starts from an extended system, for the evaluation of the higher order asymptotic terms of the center-manifold (in its normal form) only the standard linear system is required and the system extension only affects the tangent (linear) space. This aspect however is not unique to

the normal form and other parameterizations can be constructed in which the asymptotic center-manifold of a system can be evaluated using just the standard linear operator $\mathbf{L}$. Here we construct one such parameterization.

Starting again with the asymptotic representations of $\mathcal{G}(\mathbf{z})$ and $\mathcal{Y}(\mathbf{z})$ as in equations (38) and (39) respectively, one may constrain all higher order terms $\widehat{\mathbf{y}}_{i_1, \dots, i_k}$ to lie in the complement of the center subspace (spanned by $\widehat{\mathbf{V}}$), resulting in,

$$\widehat{\mathbf{w}}_j^\dagger \widehat{\mathbf{y}}_{i_1, \dots, i_k} = \mathbf{0}, \quad \forall j \in 1, \dots, m, \quad (69)$$

for all $(i_1, \dots, i_k) \in \mathcal{I}\{(k), [1, m]\}$, $k \geq 2$. Note that this condition applies regardless of resonance.

At $\mathcal{O}(|\mathbf{z}|)$ one again obtains equation (42) and, as in the case of the normal form, the unknowns $\widehat{\mathbf{y}}_{i_1}, \mathbf{g}_{i_1}$ etc., can be associated with the generalized eigenvalue problem resulting in equation (43). At first order the expressions are identical to those obtained for the normal form, which is unsurprising since the $\mathcal{O}(|\mathbf{z}|)$ terms only represent the center subspace.

Since the linear part of the reduced representation is given by $\widehat{\mathbf{K}}$, one again obtains a lower triangular system of equations for the unknown vectors $\widehat{\mathbf{y}}_{i_1, \dots, i_k}$ at all asymptotic orders, which can be solved sequentially one vector at a time. One may therefore represent the general expression for the evaluation of higher order terms as,

$$[\widehat{\mathcal{R}}(\omega_{i_1, \dots, i_k})]^{-1} \widehat{\mathbf{y}}_{i_1, \dots, i_k} + \sum_{j=1}^m g_{i_1, \dots, i_k}^j \widehat{\mathbf{v}}_j \quad (70a)$$

$$= \widehat{\mathbf{h}}_{i_1, \dots, i_k} - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^O - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^L.$$

$$\omega_{i_1, \dots, i_k} = \widehat{K}_{i_1, i_1} + \widehat{K}_{i_2, i_2} + \dots + \widehat{K}_{i_k, i_k}. \quad (70b)$$

Since the parameterization is defined by equation (69), one obtains the definition of the asymptotic terms of $\mathcal{G}$ as,

$$g_{i_1, \dots, i_k}^j = \widehat{\mathbf{w}}_j^\dagger (\widehat{\mathbf{h}}_{i_1, \dots, i_k} - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^O - \widehat{\mathbf{h}}_{i_1, \dots, i_k}^L), \quad (71)$$

$\forall j \in 1, \dots, m$.

The extended variables vanish for $\widehat{\mathbf{h}}_{i_1, \dots, i_k}$ by construction and, following the logical arguments presented in the previous subsection, it may be verified that the extended variables also vanish for $\widehat{\mathbf{h}}_{i_1, \dots, i_k}^O, \widehat{\mathbf{h}}_{i_1, \dots, i_k}^L$ and $\widehat{\mathbf{y}}_{i_1, \dots, i_k}$ (with $k \geq 2$). This

then implies that,

$$g_{i_1, \dots, i_k}^j = 0, \quad \forall j \in n+1, \dots, m. \quad (72)$$

Equation (70) then reduces to its single component form,

$$[\mathcal{R}(\omega_{i_1, \dots, i_k})]^{-1} \mathbf{y}_{i_1, \dots, i_k} + \sum_{j=1}^n g_{i_1, \dots, i_k}^j \mathbf{v}_j \quad (73a)$$

$$= \mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^O - \mathbf{h}_{i_1, \dots, i_k}^L. \quad (73b)$$

$$\omega_{i_1, \dots, i_k} = \hat{K}_{i_1, i_1} + \hat{K}_{i_2, i_2} + \dots + \hat{K}_{i_k, i_k}.$$

The simplified mappings between the extended variables and the reduced coordinate representations obtained in equations (67) and (68) apply for this particular graph representation as well.

One may define the projection operator on to the complement of the center subspace of $\mathbf{L}$ as,

$$\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger. \quad (74)$$

By construction we have, $\mathbf{Q}\mathbf{y}_{i_1, \dots, i_k} = \mathbf{y}_{i_1, \dots, i_k}$ and, $\mathbf{Q} \sum_{j=1}^n g_{i_1, \dots, i_k}^j \mathbf{v}_j = \mathbf{0}$. Expressing the resolvent operator restricted to the complement of the center subspace as

$$\mathcal{R}_s(\omega) = [\mathbf{Q}(\omega\mathbf{I} - \mathbf{L})\mathbf{Q}]^{-1} \quad (75)$$

Equation (73) can then be rewritten in a restricted subspace as,

$$[\mathcal{R}_s(\omega_{i_1, \dots, i_k})]^{-1} \mathbf{y}_{i_1, \dots, i_k} = \mathbf{Q} [\mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^O - \mathbf{h}_{i_1, \dots, i_k}^L], \quad (76a)$$

$$\omega_{i_1, \dots, i_k} = \hat{K}_{i_1, i_1} + \hat{K}_{i_2, i_2} + \dots + \hat{K}_{i_k, i_k}. \quad (76b)$$

with the asymptotic components of $\mathcal{G}$ defined as

$$g_{i_1, \dots, i_k}^j = \mathbf{w}_j^\dagger (\mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^O - \mathbf{h}_{i_1, \dots, i_k}^L), \quad (77a)$$

for $j \in 1, \dots, n$,

$$g_{i_1, \dots, i_k}^j = 0, \quad (77b)$$

for $j \in n+1, \dots, m$.

This particular graph style parameterization has been utilized by Negi [35] for the Navier–Stokes system where the Reynolds number perturbations

have been used for the system extension. The procedure for obtaining the graph representation is summarized in algorithm 4.

Algorithm 4 Center–Manifold: Graph parameterization

Require: System: $\partial \mathbf{u} / \partial t = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.

Ensure: $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $\partial \mathcal{N} / \partial \mathbf{u} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger \mathbf{L} \mathbf{V} = \Lambda_C$.

/ Center subspace dimension: n . */*

Require: Extension via $\boldsymbol{\nu}, \boldsymbol{\theta}$ and $\mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta})$.

Evaluate $\hat{\mathbf{V}}, \hat{\mathbf{K}}$ via algorithm 1.

/ Center subspace dim.: $m = n + p + h$. */*

$\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger$ */* Subspace projector. */*

Require: Maximum Asymptotic order: k_{max}

for ($k \in 2, 3, \dots, k_{max}$) **do**

for ($i_1, \dots, i_k \in \mathcal{I}\{(k), [1, m]\}$) **do**

/ Build all r.h.s. contributions. */*

$\mathbf{r} = \mathbf{h}_{i_1, \dots, i_k} - \mathbf{h}_{i_1, \dots, i_k}^O - \mathbf{h}_{i_1, \dots, i_k}^L$

$\omega = \hat{K}_{i_1, i_1} + \hat{K}_{i_2, i_2} + \dots + \hat{K}_{i_k, i_k}$

/ Build $\mathbf{g}_{i_1, \dots, i_k}$, Eq. (77) */*

for ($j \in 1, \dots, n$) **do**

$| \quad g_{i_1, \dots, i_k}^j = \mathbf{w}_j^\dagger \mathbf{r}$

end for

for ($j \in n+1, \dots, m$) **do**

$| \quad g_{i_1, \dots, i_k}^j = 0$

end for

/ Evaluate restricted resolvent */*

$\mathbf{y}_{i_1, \dots, i_k} = \mathcal{R}_s(\omega) \mathbf{Q} \mathbf{r}$, */* Eq. (76) */*

with, $\mathbf{Q} \mathbf{y}_{i_1, \dots, i_k} = \mathbf{0}$.

/ $\mathcal{R}_s(\omega) = [\mathbf{Q}(\omega\mathbf{I} - \mathbf{L})\mathbf{Q}]^{-1}$ */*

$\hat{\mathbf{y}}_{i_1, \dots, i_k} = [\mathbf{y}_{i_1, \dots, i_k}; \mathbf{0}; \mathbf{0}]$

end for

end for

return $\mathcal{G}(\mathbf{z}), \mathcal{Y}(\mathbf{z})$.

The resulting expressions for the normal form given by equations (62), (64) and, the graph style parameterization given by equations (76), (77) are quite similar. The similarity is due to the fact that the tangent space in both cases is identical, with $\hat{\mathbf{K}}$ representing the linear part of the parameterization $\mathcal{G}(\mathbf{z})$. Typically one is interested in the normal form of the reduced equations since

they result in the “simplest form” of the representation. The particular graph parameterization however is advantageous when one needs to consider the mapping from the full system state to the reduced manifold, *i.e.*, $\hat{\mathbf{u}} \mapsto \mathbf{z}$. For both normal form and the graph parameterization, mapping that is explicitly constructed is $\mathcal{Y} : \mathbf{z} \mapsto \hat{\mathbf{u}}$. The inverse map $\mathcal{Y}^{-1} : \hat{\mathbf{u}} \mapsto \mathbf{z}$ is not constructed. It is also difficult to construct from the knowledge of $\mathcal{Y}$ itself because $\mathcal{Y}$ is constructed asymptotically and is hardly ever evaluated beyond a few orders. However, since the graph parameterization is constructed with the property that higher order terms lie in the complement of the tangent space, the inverse map $\mathcal{Y}^{-1}$ can be easily constructed by leveraging this property. Therefore, recalling that the first order vectors $\hat{\mathbf{y}}_i = \hat{\mathbf{v}}_i$, we have,

$$\begin{aligned}\hat{\mathbf{u}} &= \mathcal{Y}(\mathbf{z}) = \sum_{(1)}^{[1,m]} \hat{\mathbf{y}}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \hat{\mathbf{y}}_{i_1, i_2} z_{i_1} z_{i_2} \\ &\quad + \sum_{(3)}^{[1,m]} \hat{\mathbf{y}}_{i_1, i_2, i_3} z_{i_1} z_{i_2} z_{i_3} + \dots, \\ \Rightarrow \widehat{\mathbf{W}}^\dagger \hat{\mathbf{u}} &= \sum_{(1)}^{[1,m]} \widehat{\mathbf{W}}^\dagger \hat{\mathbf{v}}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \widehat{\mathbf{W}}^\dagger \hat{\mathbf{y}}_{i_1, i_2} z_{i_1} z_{i_2} \\ &\quad + \sum_{(3)}^{[1,m]} \widehat{\mathbf{W}}^\dagger \hat{\mathbf{y}}_{i_1, i_2, i_3} z_{i_1} z_{i_2} z_{i_3} + \dots, \\ \Rightarrow \widehat{\mathbf{W}}^\dagger \hat{\mathbf{u}} &= \widehat{\mathbf{I}}\mathbf{z} + \mathbf{0} + \mathbf{0} + \dots\end{aligned}$$

Therefore, when considering the graph parameterization presented earlier, the inverse map $\mathcal{Y}^{-1} : \hat{\mathbf{u}} \mapsto \mathbf{z}$ is simply,

$$\mathbf{z} = \mathcal{Y}^{-1}(\hat{\mathbf{u}}) = \widehat{\mathbf{W}}^\dagger \hat{\mathbf{u}}. \quad (78)$$

For the normal form, one does not get this simple inverse map since the higher order terms generically do not lie in the complement of the tangent space ($\widehat{\mathbf{V}}$).

5 A Model Problem

The asymptotic center-manifold derived in section 4 is applied to a Ginzburg-Landau model described by a complex-conjugate pair of

equations as,

$$\frac{\partial A}{\partial t} = [\delta_1 \nabla + \delta_2 + \delta_3 x + \delta_4 \nabla^2] A + \delta_5 |A|^2 A, \quad (79a)$$

$$\frac{\partial A^*}{\partial t} = [\delta_1^* \nabla + \delta_2^* + \delta_3^* x + \delta_4^* \nabla^2] A^* + \delta_5^* |A|^2 A^*. \quad (79b)$$

Where, x is the spatial coordinate, ∇ and ∇^2 are the one-dimensional gradient and Laplacian operators, and the superscript $*$ represents complex conjugation. Typically such a model would arise as the simplified dynamics of a more detailed dynamical system. Here the Ginzburg-Landau model itself is treated as the underlying dynamical model, which represents the standard system in our context. Such models have often been utilized to investigate local and global bifurcations in spatially developing flows [36, 37]. It is easily seen that $A(x) = 0$ represents a fixed point of the system, with the linear operators given by

$$\begin{aligned}\mathcal{L}_0 &= [\delta_1 \nabla + \delta_2 + \delta_3 x + \delta_4 \nabla^2], \\ \mathcal{L}_0^* &= [\delta_1^* \nabla + \delta_2^* + \delta_3^* x + \delta_4^* \nabla^2].\end{aligned}$$

For semi-infinite domains, $x \in [0, \infty)$, the spectrum for $\mathcal{L}_0$ is known analytically [36] and is given by,

$$\omega_j = \delta_2 - \frac{\delta_1^2}{4\delta_4} + (\delta_4 \delta_3^2)^{1/3} \zeta_j, \quad (80)$$

and the corresponding complex conjugate for $\mathcal{L}_0^*$. Here ζ_j denotes the countable set of zeros of the Airy function. The system parameters are chosen *ad hoc* and are given in table 1, which results in a single critical eigenmode with eigenvalue $\omega_1 = 1.0\iota$ for $\mathcal{L}_0$ and its conjugate, -1.0ι for $\mathcal{L}_0^*$. All other eigenvalues have negative real parts. The lin-

Parameter	Value	Parameter	Value
δ_1^b	-0.1	δ_4^b	$(1.0 - \iota)/\sqrt{2}$
δ_2^b	$0.741 + 1.025\iota$	δ_5^b	$-0.1 + 0.1\iota$
δ_3^b	-0.125	ω_1	$+1.0\iota$

Table 1: Parameters for the Ginzburg-Landau system at bifurcation.

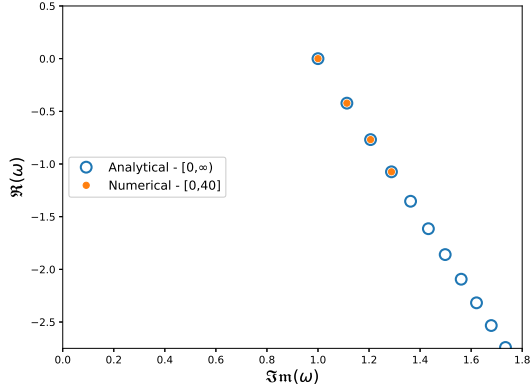


Fig. 3: Comparison of the numerically evaluated spectrum of the Ginzburg-Landau model ($\mathcal{L}_0$) in the truncated domain with the analytical spectrum in the semi-infinite domain.

ear operator at the bifurcation point for the state vector, $\mathbf{u} = [A; A^*]$ (with $\mathbf{u} \in \mathbb{C}^N$), is written as

$$\mathbf{L} = \begin{bmatrix} \mathcal{L}_0 & 0 \\ 0 & \mathcal{L}_0^* \end{bmatrix}, \quad (81)$$

which, in our context, represents the linear operator for the standard system.

A center-manifold approximation of the model problem requires the evaluation of restricted resolvents, which is a difficult problem to solve analytically. Therefore the problem is considered in a truncated domain of $x \in [0, L]$ with the boundary conditions of $A = 0$ at $x = 0$ and, $\partial A / \partial x = 0$ (natural boundary conditions) at $x = L$. Here the domain length is specified as $L = 40$. Due to the spatially localized nature of the first few eigenvectors (which dictate the global dynamics), this truncation does not affect the spectrum in any significant manner. This can be verified from the comparison of the numerically evaluated spectrum (via the implicitly restarted Arnoldi method [38]) in the truncated domain to the analytical spectrum in the semi-infinite domain (figure 3).

The standard system then has $n = 2$ critical eigenvalues and, the matrices for the critical eigenvalue and eigenvectors are given as,

$$\mathbf{\Lambda}_C = \begin{bmatrix} \iota & 0 \\ 0 & -\iota \end{bmatrix}; \quad \mathbf{V} = [\mathbf{v}_1; \mathbf{v}_2]; \quad \mathbf{W} = [\mathbf{w}_1; \mathbf{w}_2],$$

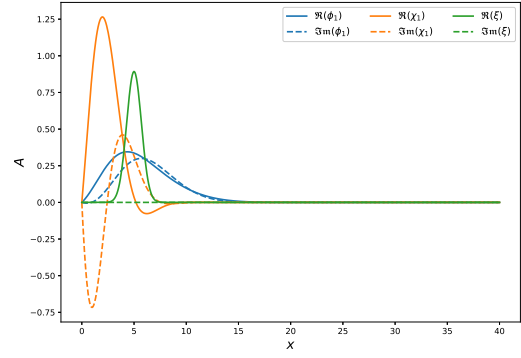


Fig. 4: The numerically evaluated eigenvector ϕ_1 , the adjoint eigenvector χ_1 and, the shape of the harmonic forcing ξ .

where the eigenvectors have the following form in terms of the two components,

$$\begin{aligned} \mathbf{v}_1 &= [\phi_1; \mathbf{0}]; & \mathbf{v}_2 &= [\mathbf{0}; \phi_1^*], \\ \mathbf{w}_1 &= [\chi_1; \mathbf{0}]; & \mathbf{w}_2 &= [\mathbf{0}; \chi_1^*], \end{aligned}$$

The numerically calculated eigenvectors ϕ_1 and χ_1 are shown in figure 4.

The system in equation (79) has a total of five parameters (ten including the complex-conjugate pairs) that can be varied from their bifurcation values. In order to avoid an explosion in the number of terms T_m^k arising from the combinatorial problem, only the variations in δ_4 and δ_4^* are considered. This suffices to illustrate the construction of the asymptotic center-manifold under parameter variations. Although the parameters are complex-conjugate pairs, it turns out to be simplest to treat them as independent parameters and enforce the complex-conjugate property later when parameter values are prescribed, which is essentially the same as treating A and A^* independently and having two independent critical eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. The extended variable for the parameter evolution is then $\boldsymbol{\nu} = [\delta_4'; \delta_4'^*]$, with

$$\delta_4 = \delta_4^b + \delta_4', \quad \delta_4^* = (\delta_4^b)^* + \delta_4'^*.$$

Harmonic forcings will be considered to illustrate the two scenarios - resonant and non-resonant external forcings. In all cases, the shape

of the external forcing is specified based on,

$$\xi(x) = \frac{1}{\beta} e^{-(x-5.0)^2}, \quad (82a)$$

$$\beta = \left[\int_0^L |e^{-(x-5.0)^2}|^2 dx \right]^{1/2}. \quad (82b)$$

The forcing shape is also shown in figure 4. In principle all cases of parameter perturbations and harmonic forcings can be considered together however, in order to keep the discussion and the number of combinatorial terms manageable, they are treated as individual cases.

5.1 Case I: Parameter Perturbations

First we consider the case of a pair of complex conjugate parameter perturbations δ'_4, δ_4^* without any external forcings. This results in a nonlinear system given by,

$$\begin{aligned} \frac{\partial A}{\partial t} &= \mathcal{L}_0 A + \delta'_4 \nabla^2 A + \delta_5^b (A^* A) A, \\ \frac{\partial A^*}{\partial t} &= \mathcal{L}_0^* A^* + \delta_4'^* \nabla^2 A^* + (\delta_5^b)^* (A^* A) A^*. \end{aligned}$$

Treating δ'_4 and $\delta_4'^*$ as dynamic variables results in an extended system with $n = 2, p = 2, h = 0$, and therefore a total critical subspace dimension of $m = 4$. The eigenvalue matrices are given by,

$$\mathbf{\Lambda}_C = \begin{bmatrix} \iota & 0 \\ 0 & -\iota \end{bmatrix}, \quad \mathbf{\Lambda}_P = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The parameter eigenvalues are non-resonant and therefore the coupling matrix $\mathbf{\Gamma}_P$ is trivial, *i.e.* ,

$$\mathbf{\Gamma}_P = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

In this particular case since the equilibrium state at the bifurcation point is trivial ($A = \mathbf{0}$), the linear matrix $\mathbf{L}_\nu$ which couples A and A^* to the extended variables is also a null matrix. Therefore, through the application of algorithm 1, the critical subspace is trivially extended as,

$$\begin{aligned} (\lambda_1^P - \mathbf{L}) \mathbf{v}_1^P &= \mathbf{0}, \\ (\lambda_2^P - \mathbf{L}) \mathbf{v}_2^P &= \mathbf{0}, \\ \mathbf{V}_P &= [\mathbf{v}_1^P, \mathbf{v}_2^P], \end{aligned}$$

resulting in the final extended matrices as,

$$\hat{\mathbf{K}} = \begin{bmatrix} \mathbf{\Lambda}_C & \mathbf{\Gamma}_P \\ \mathbf{0} & \mathbf{\Lambda}_P \end{bmatrix}, \quad \hat{\mathbf{V}} = \begin{bmatrix} \mathbf{V} & \mathbf{V}_P \\ \mathbf{0} & \mathbf{I}_p \end{bmatrix},$$

where, both $\mathbf{\Gamma}_P$ and $\mathbf{V}_P$ are null matrices (of appropriate sizes).

Once $\hat{\mathbf{V}}$ and $\hat{\mathbf{K}}$ are known, algorithm 3 may now be applied to the extended system to obtain the center-manifold approximation to the desired order. Since $\mathbf{\Gamma}_P$ is a trivial matrix $\hat{\mathbf{K}}$ is diagonal and, all asymptotic vectors $\mathbf{y}_{i_1, \dots, i_k}$ can be evaluated independent of each other. For $m = 4$, one obtains $T_4^2 = 10$ terms at second order and $T_4^3 = 20$ terms at third order. The final reduced equations (reduced representation $\mathcal{G}(\mathbf{z})$) are found to be,

$$\begin{aligned} \frac{dz_1}{dt} &= +\iota z_1 - (0.169 - 0.153\iota) z_1 z_3 \\ &\quad - (0.008 - 0.014\iota) z_1 z_1 z_2 \\ &\quad + (0.194 - 0.114\iota) z_1 z_3 z_3, \\ \frac{dz_3}{dt} &= 0, \end{aligned}$$

along with the complex conjugated equations for z_2 and z_4 . One may substitute $z_1 \mapsto z$, $z_2 \mapsto z^*$, $z_3 \mapsto \delta'_4$, $z_4 \mapsto \delta_4'^*$, to obtain expressions directly relating mode amplitude evolution to the parameter perturbations as,

$$\begin{aligned} \frac{dz}{dt} &= +\iota z - (0.169 - 0.153\iota) \delta_4'^* z \\ &\quad - (0.008 - 0.014\iota) |z|^2 z \\ &\quad + (0.194 - 0.114\iota) (\delta_4')^2 z. \end{aligned} \quad (84)$$

The results of the asymptotic center-manifold approximations compared with the fully non-linear response of the Ginzburg-Landau system are shown in figure 5. The results compare very well for small δ'_4 values, with progressive loss of accuracy for larger parameter variations which is typical of asymptotic results.

A similar comparison for the angular frequency of the saturated limit-cycle is shown in figure 6. The center-manifold approximation provides a fairly accurate reproduction of the non-linear angular frequency variation.

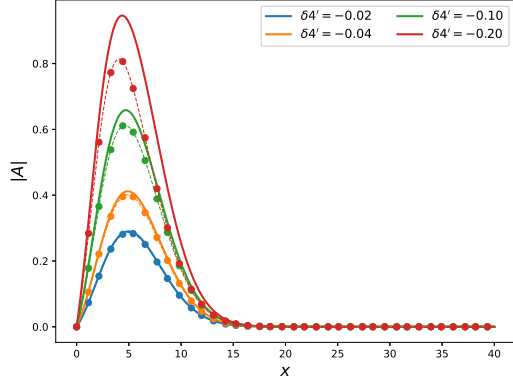


Fig. 5: Spatial shape of the system response ($|A|$) to perturbations of δ_4 parameter. The solid lines indicate results of the non-linear simulations of the Ginzburg-Landau system and the dashed lines with markers indicate the results of the center-manifold approximation.

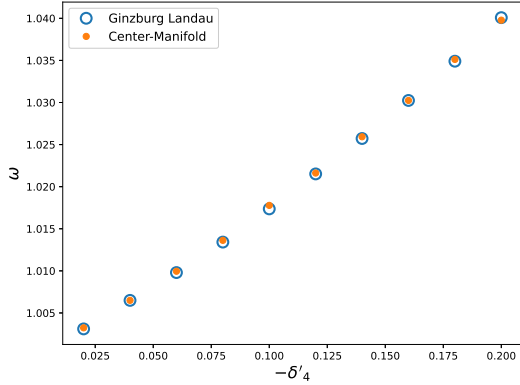


Fig. 6: Variation of the final limit-cycle-oscillation angular frequency due to perturbations of δ_4 parameter. The empty circles indicate results of the non-linear simulations of the Ginzburg-Landau system and the filled circles indicate the results of the center-manifold approximation.

5.2 Case II: Non-Resonant Forcing

For the second case the system is forced with a pair of non-resonant complex-conjugate forcings ($h = 2$) with $\lambda^H = [1.3\iota; -1.3\iota]$ and forcing shapes

f_i^H given by

$$f_1^H = \begin{bmatrix} \xi(x) \\ 0 \end{bmatrix}, \quad f_2^H = \begin{bmatrix} 0 \\ \xi^*(x) \end{bmatrix}, \quad (85)$$

where, $\xi(x)$ is defined in equation (82). The governing equations can be written as

$$\frac{\partial A}{\partial t} = \mathcal{L}_0 A + \delta_5^b(A^* A)A + \xi(x)\theta, \quad (86a)$$

$$\frac{\partial A^*}{\partial t} = \mathcal{L}_0^* A^* + (\delta_5^b)^*(A^* A)A^* + \xi^*(x)\theta^*, \quad (86b)$$

$$\frac{d\theta}{dt} = \lambda_1^H \theta, \quad (86c)$$

$$\frac{d\theta^*}{dt} = (\lambda_1^H \theta)^*. \quad (86d)$$

No parameter perturbations are considered, resulting in $n = 2$, $p = 0$, $h = 2$ and $m = 4$. Since the external forcing is non-resonant, one obtains the following matrices,

$$\Lambda_C = \begin{bmatrix} \iota & 0 \\ 0 & -\iota \end{bmatrix}, \Lambda_H = \begin{bmatrix} 1.3\iota & 0 \\ 0 & -1.3\iota \end{bmatrix}, \quad \Gamma_H = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The forcing modes are non-trivial and are evaluated as follows,

$$\begin{aligned} (\lambda_1^H - \mathbf{L})\mathbf{v}_1^H &= \mathbf{f}_1^H, \\ (\lambda_2^H - \mathbf{L})\mathbf{v}_2^H &= \mathbf{f}_2^H, \\ \mathbf{V}_H &= [\mathbf{v}_1^H, \mathbf{v}_2^H]. \end{aligned}$$

The resulting extended matrices are given by

$$\hat{\mathbf{K}} = \begin{bmatrix} \Lambda_C & \Gamma_H \\ 0 & \Lambda_H \end{bmatrix}, \quad \hat{\mathbf{V}} = \begin{bmatrix} \mathbf{V} & \mathbf{V}_H \\ 0 & \mathbf{I}_h \end{bmatrix},$$

where Γ_H is a trivial matrix and therefore $\hat{\mathbf{K}}$ is diagonal.

As done in section 5.1, the application of algorithm 3 results in the evolution equations in their normal form for the reduced system,

$$\begin{aligned} \frac{dz}{dt} &= +\iota z \\ &\quad - (0.008 - 0.014\iota)|z|^2 z \\ &\quad - (0.026 - 0.051\iota)|\theta|^2 z, \\ \frac{d\theta}{dt} &= +1.3\iota\theta. \end{aligned} \quad (87)$$

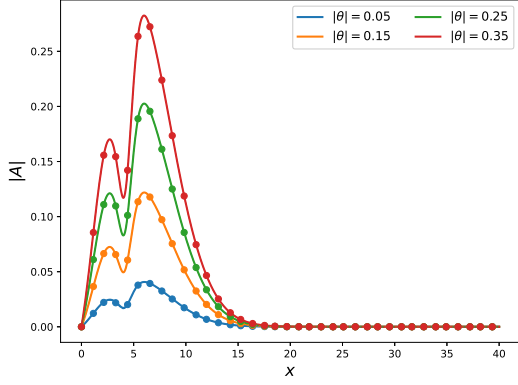


Fig. 7: Spatial shape of the system response ($|A|$) to harmonic forcing with amplitude $|\theta|$. The solid lines indicate results of the non-linear simulations of the Ginzburg-Landau system and the dashed lines with markers indicate the results of the center-manifold approximation.

Here, the substitutions $z_1 \mapsto z$, $z_2 \mapsto z^*$, $z_3 \mapsto \theta$ and $z_4 \mapsto \theta^*$, have already been performed. One can observe that the amplitude evolution of the original critical mode, denoted by z , only depends on the forcing amplitude $|\theta|$ and not on its instantaneous phase. The comparison of the response of the reduced system to that of the fully non-linear system is shown in figure 7. The response curves are nearly identical.

Since $|\theta|$ is an a priori known quantity which remains invariant in time, it can simply be treated as another constant parameter and, the effect of non-resonant harmonic forcing on the critical mode of the system can easily be understood by looking at the linear part of the reduced system, *i.e.* $[-0.0258|\theta|^2 + (1 + 0.0507|\theta|^2)\iota]z$. The harmonic forcing effectively increases the angular frequency, but also induces a negative growth rate, and therefore a stabilizing effect on the original critical mode. The net result is that the original critical mode eventually decays to zero and vanishes, even if it may be initially excited, and the final response is simply the forced response to the harmonic forcing. Of course this may not always be the case and the forcing can induce a destabilizing behavior of the original critical mode.

5.3 Case III: Resonant Forcing

As in the previous subsection we again consider the case of the system forced with a pair of complex conjugate harmonic forcings. This time however the angular frequencies of the external forcings are resonant with the internal eigenmodes of the system. Accordingly, the eigenvalues of the forcing modes is given by $\lambda^H = [\iota; -\iota]$ and the eigenvalue matrices are given by,

$$\Lambda_C = \begin{bmatrix} \iota & 0 \\ 0 & -\iota \end{bmatrix}, \quad \Lambda_H = \begin{bmatrix} \iota & 0 \\ 0 & -\iota \end{bmatrix}$$

The forcing shapes are again given by equation (85) and the resulting forced system is again described by equation (86). Due to resonance, the matrix Γ_H is no longer trivial and its individual entries are evaluated according to algorithm 1 as,

$$\begin{aligned} (\gamma_1^H)_1 &= \mathbf{w}_1^\dagger \mathbf{f}_1^H, & (\gamma_2^H)_1 &= 0, \\ (\gamma_1^H)_2 &= 0, & (\gamma_2^H)_2 &= \mathbf{w}_2^\dagger \mathbf{f}_2^H, \end{aligned}$$

resulting in,

$$\Gamma_H = [\gamma_1^H, \gamma_2^H] = \begin{bmatrix} 0.151 - 0.481\iota & 0 \\ 0 & 0.151 + 0.481\iota \end{bmatrix}.$$

The forcing modes for the extended tangent space are evaluated as,

$$\begin{aligned} (\lambda_1^H - \mathbf{L})\mathbf{v}_1^H &= \mathbf{f}_1^H - \mathbf{V}\gamma_1^H, \\ (\lambda_2^H - \mathbf{L})\mathbf{v}_2^H &= \mathbf{f}_2^H - \mathbf{V}\gamma_2^H, \\ \mathbf{V}_H &= [\mathbf{v}_1^H, \mathbf{v}_2^H]. \end{aligned}$$

The extended matrices being,

$$\hat{\mathbf{K}} = \begin{bmatrix} \Lambda_C & \Gamma_H \\ \mathbf{0} & \Lambda_H \end{bmatrix}, \quad \hat{\mathbf{V}} = \begin{bmatrix} \mathbf{V} & \mathbf{V}_H \\ \mathbf{0} & \mathbf{I}_h \end{bmatrix},$$

where, Γ_H is no longer trivial and $\hat{\mathbf{K}}$ has an upper triangular structure.

The asymptotic center-manifold evaluation through the application of algorithm 3 results in

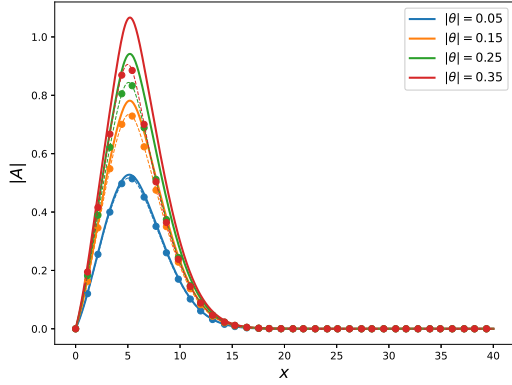


Fig. 8: Spatial shape of the system response ($|A|$) to resonant harmonic forcing with amplitude $|\theta|$. The solid lines indicate results of the non-linear simulations of the Ginzburg-Landau system and the dashed lines with markers indicate the results of the center-manifold approximation.

the following reduced system,

$$\begin{aligned}
 \frac{dz}{dt} = & + \iota z + (0.151 - 0.482\iota)\theta \\
 & - (0.008 - 0.014\iota)|z|^2 z \\
 & + (0.012 + 0.019\iota)z^2 \theta^* \\
 & - (0.012 - 0.020\iota)|z|^2 \theta \\
 & + (0.001 + 0.070\iota)|\theta|^2 z \\
 & - (0.013 - 0.031\iota)z^* \theta^2 \\
 & - (0.004 - 0.067\iota)|\theta|^2 \theta \\
 \frac{d\theta}{dt} = & + \iota \theta,
 \end{aligned} \tag{88}$$

and its conjugate for z^* and θ^* . Clearly, in the case of resonant forcings, the original critical mode evolution depends on the amplitude as well as the phase of the external forcing.

A comparison of the system response between the non-linear results and the center-manifold approximation is shown in figure 8. Again a good agreement is found for small values of $|\theta|$ with systematic deviation for larger amplitudes typical of asymptotic methods.

As a final note, one may observe that performing the substitutions, $\delta'_4 \mapsto 0$ in equation (84) and, $\theta \mapsto 0$ in equations (87) and (88), reduces all three

equations to,

$$\frac{dz}{dt} = + \iota z - (0.008 - 0.014\iota)|z|^2 z. \tag{89}$$

This is the normal form of the center-manifold for the standard Ginzburg-Landau system (79). This provides further numerical validation of the fact that the center-manifold approximation of the extended system reduces to that of the standard system whenever the system perturbation and harmonic forcings vanish.

6 Discussion and Summary

Perhaps the most useful aspect of considering system extension is that it elegantly unifies both parameter perturbations and application of harmonic forcings into a single framework. When viewed through the lens of an extended system a parameter perturbation is nothing but a harmonically forced system, albeit with a zero angular frequency (and nonlinearities). It easily extends to cases where the system parameters themselves have an oscillatory variation, which often occurs in engineering applications, with oscillating boundary conditions being a prime example. In the previous sections parameter perturbations and harmonic forcings have been treated as separate cases however, this is primarily to illustrate the identical algorithmic treatment required for the extension of the tangent space in the two cases. For the extension of the center subspace the only thing that is important is whether the extended eigenvalues are in resonance with the intrinsic critical eigenvalues of the system. For the non-resonant cases the subspace extends in a straightforward manner, while in the case of resonance one obtains a Jordan form like structure with a generalized eigenvector.

Secondly, the system extension also illuminates the structure of the problem under forcing or parameter perturbation, with the extended directions effectively being the resolvent responses of the system. Of prime importance is the evaluation of the adjoint extended modes of the system that are essential in constructing the projection operators for the individual subspace directions. The extended adjoint modes have a special structure regardless of resonance, which was fruitfully utilized to show that, when constructing asymptotic

center-manifold approximations of the extended system, all evaluations of the asymptotic vectors, $\hat{\mathbf{y}}_{i_1, \dots, i_k}$, in fact only require the original linear system $\mathbf{L}$. This is of significant importance when considering approximations for large systems where specialized algorithms are often tailored to exploit the inherent structure of $\mathbf{L}$. Despite the system extension, only the original linear system $\mathbf{L}$ is required for the calculations, which means that the specialized algorithms do not require any substantial modifications. In that sense the current approach provides the best of both worlds. It allows the calculation of center-manifold approximations of perturbed systems using the original unperturbed linear operators. At the same time it provides the theoretical perspective of understanding the perturbations as intrinsic modes of an extended system.

It is also shown that the $\hat{\mathbf{K}}$ matrix, which is the linear part of the reduced representation $\mathcal{G}(\mathbf{z})$, is responsible for the cross coupling of vectors $\mathbf{y}_{i_1, \dots, i_k}$ at each order k in the asymptotic center-manifold approximations. However, this matrix is upper triangular, at least in the normal form and graph-method approximations described in sections 4.1 and 4.3. The consequence of the upper triangular nature of $\hat{\mathbf{K}}$ is that the linear system of equations obtained at each order k for the solution of the asymptotic vectors $\mathbf{y}_{i_1, \dots, i_k}$ is lower triangular. Therefore each vector can be solved for, one at a time, sequentially using back substitution. For large systems this is an important property since the coupled system solution can quickly become computationally infeasible.

The derived center-manifold approximation is applied to a model Ginzburg-Landau problem at the bifurcation point. The system is considered with perturbations of the diffusion coefficient (δ_4) as well as with resonant and non-resonant external forcings. In all cases a very good agreement is found between the non-linear system response and the response predicted by the calculated center-manifold approximations.

Finally, one may note that in the current study, only linear forcing was considered, so that $\mathbf{f}^H(\boldsymbol{\theta})$ defined in equation (8) only has terms with linear dependence on $\boldsymbol{\theta}$. However, this term may easily be modified to include nonlinear terms of the type $\boldsymbol{\theta}\boldsymbol{\theta}$, $\boldsymbol{\theta}\mathbf{u}$, etc.. This makes no difference in the overall procedure except that the nonlinear

terms $\mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta})$ need to include the added nonlinearities. The rest of the procedure for tangent space extension and center-manifold evaluation follows exactly as before. In fact, since we now understand that $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$ can be treated equivalently, the parameter perturbation term $\delta'_4 \nabla^2 A$ is precisely one such term. If one endows δ'_4 with an oscillatory variation one obtains what Thurnher et al. [28] refer to as parametric forcing. From the point of view of the extended system $\mathfrak{Im}(\lambda_i^P)$ being non-zero makes no algorithmic difference.

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